A Sharp Berry–Esseen Theorem and First Edgeworth Expansion for the Erdős–Turán Law of Comparable-Rate Luce Permutations

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Research supplement

Abstract

Let σ_n be the Luce permutation induced by independent exponential clocks whose rates, after a common rescaling, lie in $[1, \kappa]$ for one fixed $\kappa < \infty$, and put $L = \log n$. The companion paper proves the Erdős–Turán central limit theorem for $\log \operatorname{ord}(\sigma_n)$. This quantitative supplement proves the natural Berry–Esseen rate and the first Edgeworth expansion for the actual permutation order.

The original switching characteristic-function argument first gives the sharp proxy estimate

$$\mathrm{d_K}\left(\frac{B_n^\circ - \mu_n}{\sqrt{v_n}}, N(0,1)\right) = O_\kappa(L^{-1/2}).$$

A fixed-order multi-clock identity also yields all fixed moments of the product–least-common-multiple remainder and, independently, the almost-sharp family $O_{\kappa,\varepsilon}(L^{-1/2+\varepsilon})$.

For the endpoint theorem we use four additional modules. A free-depth finite-endpoint theorem, specialized to terminal depth $m \asymp \log(1/\delta)$, gives polynomial product–Poisson approximation of the complete intermediate cycle vector, jointly with passive marks. A prime-power linearization replaces the small-prime part of the nonlinear least-common-multiple correction by a linear cycle statistic. The resulting independent compound-Poisson statistic admits a first Edgeworth expansion; an extended logarithmic Fourier window reduces the smoothing remainder to $O((\log L)^2/L)$. Finally, Poisson Poincaré, fixed-order graph estimates, and a residual-gap argument control the centered submacroscopic and macroscopic arithmetic remainders.

Writing $O_n = \operatorname{ord}(\sigma_n)$ and $m_n = \mathbb{E} \log O_n$, we prove

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{L^3/3}}, N(0,1)\right) \le \frac{C_\kappa}{\sqrt{L}},$$

and, uniformly in x,

$$\mathbb{P}\left\{\frac{\log O_n - m_n}{\sqrt{L^3/3}} \le x\right\} = \Phi(x) - \frac{\sqrt{3}}{8}(x^2-1)\varphi(x)L^{-1/2} + o_\kappa(L^{-1/2}).$$

Consequently the order $L^{-1/2}$ is optimal. The exact mean is essential at this precision; an explicit deterministic center accurate to $o(L)$ is not supplied here.

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1 Introduction and theorem map

For a permutation $\pi \in S_n$,

$$\operatorname{ord}(\pi) = \operatorname{lcm}\{\text{cycle lengths of } \pi\}.$$

The classical theorem of Erdős and Turán states that the logarithm of the order of a uniform random permutation is asymptotically Gaussian on the $(\log n)^{3/2}$ scale [4]. Barbour and Tavaré

proved a sharp rate theorem in the uniform setting [3]; related quantitative ideas belong to the theory of logarithmic combinatorial structures [1].

The companion paper [6] establishes the qualitative Erdős–Turán law for every comparable-rate Luce triangular array: if, after a common rescaling,

$$1 \le \theta_{n,i} \le \kappa,$$

then

$$\frac{\log \mathrm{ord}(\sigma_n) - \frac{1}{2}(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}} \Rightarrow N(0,1).$$

The present supplement quantifies that law through the first non-Gaussian term.

Throughout, put

$$L = \log n, \qquad b_n = \lceil L^3 \rceil, \qquad U_n = \left\lfloor \frac{n}{L^{A_0}} \right\rfloor,$$

where A_0 is chosen sufficiently large in terms of κ and the polynomial accuracy required below. For $b_n < r \le U_n$, write

$$a_{n,r} = \frac{n}{r(n-r)}, \qquad w_r = \log r,$$

and define

$$B_n^\circ = \sum_{b_n < r \le U_n} w_r C_{n,r},$$
$$\mu_n = \sum_{b_n < r \le U_n} a_{n,r} w_r,$$
$$v_n = \sum_{b_n < r \le U_n} a_{n,r} w_r^2.$$

Also put

$$O_n = \mathrm{ord}(\sigma_n), \qquad m_n = \mathbb{E} \log O_n, \qquad R_n = \log O_n - B_n^\circ.$$

The previously established checkpoint hierarchy is retained because it uses weaker endpoint input and supplies the fixed-order arithmetic estimates used again in the sharp proof.

Theorem 1.1 (Sharp checkpoint for the additive proxy)**.** *For every fixed $\kappa < \infty$,*

$$\mathrm{d_K}\left(\frac{B_n^\circ - \mu_n}{\sqrt{v_n}}, N(0,1) \right) \le \frac{C_\kappa}{\sqrt{L}}.$$

Theorem 1.2 (L^1 quantitative order checkpoint)**.** *One has*

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{v_n}}, N(0,1) \right) \le C_\kappa L^{-1/4} \sqrt{\log L}.$$

The same bound holds after replacing m_n and $\sqrt{v_n}$ by $\frac{1}{2}L^2$ and $\sqrt{L^3/3}$, respectively.

Theorem 1.3 (Second-moment convergence checkpoint)**.** *One has*

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{v_n}}, N(0,1) \right) \le C_\kappa L^{-1/3} (\log L)^{2/3}.$$

The same rate holds with centering $\frac{1}{2}L^2$ and scale $\sqrt{L^3/3}$.

Theorem 1.4 (All fixed moments of the order remainder)**.** *For every fixed integer $k \geq 1$, uniformly over all comparable-rate triangular arrays and for all sufficiently large n,*

$$\mathbb{E}|R_n|^k \leq C_{\kappa,k} L^k (\log L)^k.$$

Consequently,

$$\|R_n - \mathbb{E}R_n\|_k \leq C_{\kappa,k} L \log L.$$

Theorem 1.5 (Fixed-moment quantitative order laws)**.** *For every fixed integer $k \geq 1$,*

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{v_n}}, N(0,1)\right) \leq C_{\kappa,k}\left[L^{-1/2} + L^{-k/(2(k+1))}(\log L)^{k/(k+1)}\right].$$

The same bound holds after replacing m_n and $\sqrt{v_n}$ by $\frac{1}{2}L^2$ and $\sqrt{L^3/3}$, respectively.

Corollary 1.6 (Almost-sharp Berry–Esseen law)**.** *For every fixed $\varepsilon > 0$,*

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{v_n}}, N(0,1)\right) \leq C_{\kappa,\varepsilon} L^{-1/2+\varepsilon}.$$

The same rate holds with the classical leading centering and scale.

The new endpoint results are the following.

Theorem 1.7 (Polynomial passive-mark product–Poisson approximation)**.** *Let*

$$\xi_n = (C_{n,r})_{b_n < r \leq U_n}.$$

For every fixed $Q > 0$, after choosing $A_0 = A_0(\kappa, Q)$ sufficiently large,

$$\mathrm{d_{TV}}\left(\mathcal{L}(\xi_n), \bigotimes_{b_n < r \leq U_n} \mathrm{Pois}(a_{n,r})\right) \leq C_{\kappa,Q} L^{-Q}.$$

More generally the same estimate holds jointly with any standard-Borel, permutation-measurable passive mark that is unchanged by every good core split. In particular it holds jointly with

$$K_n^+ = \sum_{r > U_n} C_{n,r},$$

and the Poisson core is independent of the copied mark on the comparison space.

Theorem 1.8 (Sharp Berry–Esseen law for the order)**.** *Uniformly over all comparable-rate triangular arrays,*

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{L^3/3}}, N(0,1)\right) \leq \frac{C_\kappa}{\sqrt{L}}.$$

Theorem 1.9 (First Edgeworth expansion)**.** *Uniformly in $x \in \mathbb{R}$ and uniformly over all comparable-rate triangular arrays,*

$$\mathbb{P}\left\{\frac{\log O_n - m_n}{\sqrt{L^3/3}} \leq x\right\} = \Phi(x) - \frac{\sqrt{3}}{8}(x^2-1)\varphi(x)L^{-1/2} + o_\kappa(L^{-1/2}). \tag{1}$$

Corollary 1.10 (Optimality of the Berry–Esseen order)**.** *One has*

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{L^3/3}}, N(0,1)\right) \geq \left(\frac{\sqrt{3}}{8\sqrt{2\pi}} + o_\kappa(1)\right) L^{-1/2}.$$

Dependency map

The checkpoint hierarchy through Corollary 1.6 uses the square-root-depth switching input isolated in the original supplement. The sharp and Edgeworth theorems additionally use the free-depth finite-endpoint theorem proved in the switching manuscript [7]. Its linear-depth specialization supplies arbitrary polynomial endpoint accuracy. The remaining ingredients—the passive-mark Campbell–Stein argument, the independent Poisson Edgeworth analysis, the prime-power linearization, and the centered remainder estimates—are proved below.

2 Imported switching and arithmetic inputs

The model and notation are those of [6]. Let $X_{n,i} \sim \mathrm{Exp}(\theta_{n,i})$ be independent and let $\sigma_n(i)$ be the rank of $X_{n,i}$. Write $C_{n,r}$ for the number of r-cycles and $L_n(i)$ for the cycle length containing i. For distinct labels i, j, define the clock-swap factor

$$R_{ij} = \exp\{-(\theta_{n,i} - \theta_{n,j})(X_{n,j} - X_{n,i})\},$$

and the switching intensity

$$S_{n,r} = \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}.$$

Fix $1 < p < \lambda_\kappa$, where

$$\lambda_\kappa = \begin{cases} \infty, & \kappa = 1, \\ \kappa/(\kappa-1), & \kappa > 1, \end{cases}$$

and put $\beta = 1 - 1/p$. We choose

$$A_0 > \max\left\{32\kappa^2, \frac{4}{\beta}, 4\right\}. \tag{2}$$

This is far from optimized; it is chosen only to make all polynomial endpoint errors smaller than the reset error.

We use the following theorem-level interface from the companion paper.

Input theorem 2.1 (Quantitative switching ODE)**.** *Let*

$$Z_n(s) = \mathbb{E}e^{sB_n^\circ}, \qquad H_n(s) = \sum_{b_n<r\le U_n} a_{n,r} w_r e^{sw_r}.$$

For every s on the imaginary axis,

$$Z_n'(s) = H_n(s)Z_n(s) + E_n(s). \tag{3}$$

Moreover,

$$\sup_{\xi\in\mathbb{R}} |E_n(i\xi)| \le \sum_j \log(2a_j) \sup_{a_j\le r<2a_j} \mathbb{E}\left|\frac{S_{n,r}}{n} - 1\right| + C_\kappa L^{2-A_0\beta}, \tag{4}$$

where $a_j = 2^j b_n$ and the sum is over $a_j \le U_n$. The one-block estimate and its explicit endpoint error imply

$$\mathfrak{e}_n := \sup_{\xi\in\mathbb{R}} |E_n(i\xi)| \le C_\kappa L \exp\{-c_\kappa\sqrt{\log L}\}. \tag{5}$$

Derivation of the final bound. The proof of the weighted dyadic estimate in [6] splits the one-block error into a reset term and terms carrying positive powers of $\delta_j = 4a_j/n$, together with superpolynomially small environment failures. The reset contribution is

$$CL(1+\sqrt{\log L})e^{-c\sqrt{A_0 \log L}},$$

which is bounded by $CLe^{-c'\sqrt{\log L}}$ after changing constants. After the exponent $1/(2\kappa)$ from the truncation optimization, the smallest positive power of δ_j is at least $\delta_j^{1/(8\kappa^2)}$. The choice (2) makes the weighted sum of every such term $O(L^{-2})$. The terms containing $(na_j)^{-1/2}$ and the environment failures are smaller. Finally, $L^{2-A_0\beta} = O(L^{-2})$. This proves (5). □

We also import the arithmetic comparison from the companion paper.

Input theorem 2.2 (Truncation and product–LCM discrepancy)**.** *Let*

$$B_n = \sum_{r=1}^{n} (\log r) C_{n,r},$$

and let

$$O_n = \operatorname{ord}(\sigma_n), \qquad O_n^\circ = \operatorname{lcm}\{r : b_n < r \le U_n,\ C_{n,r} > 0\}.$$

Define

$$\Delta_n = B_n - B_n^\circ, \qquad G_n = B_n^\circ - \log O_n^\circ.$$

Then

$$\mathbb{E}\Delta_n + \mathbb{E}G_n \le C_\kappa L \log L. \tag{6}$$

If

$$R_n = \log O_n - B_n^\circ,$$

then deterministically

$$-G_n \le R_n \le \Delta_n, \qquad |R_n| \le \Delta_n + G_n. \tag{7}$$

The first input contains the entire probabilistic switching approximation needed for Theorem 1.1; the second contains the full arithmetic information used for Theorem 1.2.

3 Deterministic exponent sums and the ideal proxy

For fixed $k \ge 1$, put

$$\rho_{k,n} = \sum_{b_n < r \le U_n} a_{n,r} w_r^k.$$

Thus $\rho_{1,n} = \mu_n$ and $\rho_{2,n} = v_n$.

Lemma 3.1 (Exponent sums)**.** *For every fixed integer $k \ge 1$,*

$$\rho_{k,n} = \frac{L^{k+1}}{k+1} + O_k(L^k \log L). \tag{8}$$

In particular,

$$\mu_n = \frac{1}{2}L^2 + O(L\log L), \tag{9}$$

$$v_n = \frac{1}{3}L^3 + O(L^2\log L), \tag{10}$$

$$\rho_{3,n} = \frac{1}{4}L^4 + O(L^3\log L), \tag{11}$$

$$\rho_{4,n} = \frac{1}{5}L^5 + O(L^4\log L). \tag{12}$$

Proof. Use the exact identity

$$a_{n,r} = \frac{1}{r} + \frac{1}{n-r}.$$

Since $U_n = o(n)$,

$$\sum_{b_n < r \le U_n} \frac{(\log r)^k}{n-r} \le C\frac{U_n L^k}{n} = O(L^{k-A_0}).$$

Integral comparison gives

$$\sum_{b_n < r \le U_n} \frac{(\log r)^k}{r} = \frac{(\log U_n)^{k+1} - (\log b_n)^{k+1}}{k+1} + O_k(1).$$

Now

$$\log U_n = L - A_0\log L + O(1), \qquad \log b_n = 3\log L + O(1),$$

and expansion proves (8). □

Let $(N_{n,r})_{b_n < r \le U_n}$ be independent Poisson variables with

$$N_{n,r} \sim \mathrm{Pois}(a_{n,r}),$$

and define the ideal compound-Poisson statistic

$$Y_n^\circ = \sum_{b_n < r \le U_n} w_r N_{n,r}.$$

Its mean is μ_n, its variance is v_n, and its centered characteristic function is

$$\psi_n(t) = \exp\left\{ \sum_{b_n < r \le U_n} a_{n,r}\left(e^{itw_r/\sqrt{v_n}} - 1 - \frac{itw_r}{\sqrt{v_n}} \right) \right\}. \tag{13}$$

Proposition 3.2 (Ideal standardized cumulants)**.** *For every fixed $j \ge 3$, the jth standardized cumulant of Y_n° is*

$$\frac{\rho_{j,n}}{v_n^{j/2}} = \frac{3^{j/2}}{j+1}L^{1-j/2}(1+o(1)).$$

In particular,

$$\frac{\rho_{3,n}}{v_n^{3/2}} = \frac{3\sqrt{3}}{4}L^{-1/2}(1+o(1)),$$

$$\frac{\rho_{4,n}}{v_n^2} = \frac{9}{5}L^{-1}(1+o(1)).$$

Proof. For a Poisson variable, every cumulant equals its mean. Cumulants add over independent summands, so the jth cumulant of Y_n° is $\rho_{j,n}$. Apply Lemma 3.1. □

Remark 3.3. Proposition 3.2 identifies the natural first Edgeworth scale of the ideal proxy. It does not, by itself, prove a Kolmogorov lower bound for B_n° or for $\log O_n$, because transferring a first Edgeworth coefficient requires an error smaller than $L^{-1/2}$ and control of the arithmetic remainder at the level of third cumulants.

4 A sharp Berry–Esseen bound for the additive proxy

Let

$$\phi_n(t) = \mathbb{E}\exp\left\{\frac{it(B_n^\circ - \mu_n)}{\sqrt{v_n}}\right\}.$$

Write

$$\Psi_n(s) = \sum_{b_n < r \le U_n} a_{n,r}(e^{sw_r} - 1).$$

Then the ideal characteristic function in (13) is

$$\psi_n(t) = e^{-it\mu_n/\sqrt{v_n}} e^{\Psi_n(it/\sqrt{v_n})}.$$

We use the standard Esseen smoothing inequality in the form

$$\mathrm{d_K}(X, N(0,1)) \le C\int_{-T}^{T} \frac{|\phi_X(t) - e^{-t^2/2}|}{|t|}\,dt + \frac{C}{T}, \tag{14}$$

where the integrand at zero is interpreted by continuity; see, for example, [8, Chapter V].

Lemma 4.1 (Gaussian window for the ideal exponent)**.** *There are constants $c_0, c_1, C > 0$ such that, with*

$$T_n = c_0\sqrt{L},$$

the following hold for all sufficiently large n and $|t| \le T_n$:

$$\log|\psi_n(t)| \le -c_1 t^2, \tag{15}$$

$$|\psi_n(t) - e^{-t^2/2}| \le CL^{-1/2}|t|^3 e^{-c_1 t^2}. \tag{16}$$

Proof. Put

$$x_{n,r} = \frac{w_r}{\sqrt{v_n}}.$$

By (10),

$$\max_{r \le U_n} x_{n,r} \le \frac{C}{\sqrt{L}}.$$

Choose c_0 so small that

$$|t|x_{n,r} \le \theta_0 < \frac{\pi}{2}$$

whenever $|t| \le T_n$. On this interval, $1 - \cos y \ge cy^2$. Hence

$$\log|\psi_n(t)| = -\sum_r a_{n,r}(1 - \cos(tx_{n,r})) \le -ct^2 \sum_r a_{n,r}x_{n,r}^2 = -ct^2,$$

which proves (15).

Let

$$\chi_n(t) = \log \psi_n(t).$$

Taylor's formula and $|tx_{n,r}| \le \theta_0$ give

$$\begin{aligned}\left|\chi_n(t) + \frac{1}{2}t^2\right| &\le C|t|^3 \sum_r a_{n,r}|x_{n,r}|^3 \\ &= C|t|^3 \frac{\rho_{3,n}}{v_n^{3/2}} \le CL^{-1/2}|t|^3.\end{aligned}$$

Moreover, both $\Re\chi_n(t)$ and $-t^2/2$ are at most $-ct^2$ after possibly decreasing c. The integral identity

$$e^{\chi_n(t)} - e^{-t^2/2} = \left(\chi_n(t) + \frac{1}{2}t^2\right) \int_0^1 e^{s\chi_n(t)-(1-s)t^2/2}\, ds$$

therefore yields (16). □

Lemma 4.2 (Frequency-uniform switching comparison)**.** *For $|t| \le T_n$,*

$$|\phi_n(t) - \psi_n(t)| \le C\frac{\mathfrak{e}_n}{\sqrt{v_n}} \min\{|t|, |t|^{-1}\}, \tag{17}$$

where the right side is interpreted as zero at $t = 0$.

Proof. The ODE (3), with $Z_n(0) = 1$, gives

$$Z_n(s) = e^{\Psi_n(s)}\left[1 + \int_0^s e^{-\Psi_n(z)} E_n(z)\, dz\right].$$

Take $s = it/\sqrt{v_n}$ and, by symmetry, assume $t \ge 0$. Parameterizing $z = iu/\sqrt{v_n}$ gives

$$|\phi_n(t) - \psi_n(t)| \le \frac{\mathfrak{e}_n}{\sqrt{v_n}} \int_0^t \exp\left\{\Re\left[\Psi_n(it/\sqrt{v_n}) - \Psi_n(iu/\sqrt{v_n})\right]\right\} du.$$

For $0 \le u \le t \le T_n$, all phases lie in $[0, \theta_0]$. Since $\sin y \ge cy$ there,

$$\cos(tx) - \cos(ux) \le -c(t^2 - u^2)x^2.$$

Summing with $x = x_{n,r}$ gives

$$\Re\left[\Psi_n(it/\sqrt{v_n}) - \Psi_n(iu/\sqrt{v_n})\right] \le -c(t^2 - u^2),$$

because $\sum_r a_{n,r}x_{n,r}^2 = 1$. Thus the integral is at most t when $t \le 1$, while for $t \ge 1$,

$$\int_0^t e^{-c(t^2-u^2)}\, du \le \int_0^t e^{-ct(t-u)}\, du \le \frac{C}{t}.$$

This proves (17). □

Proof of Theorem 1.1. Apply (14) with $T = T_n$. Split

$$\phi_n(t) - e^{-t^2/2} = \big(\phi_n(t) - \psi_n(t)\big) + \big(\psi_n(t) - e^{-t^2/2}\big).$$

By Lemma 4.2,

$$\begin{aligned}\int_{-T_n}^{T_n} \frac{|\phi_n(t) - \psi_n(t)|}{|t|}\, dt &\le C\frac{\mathfrak{e}_n}{\sqrt{v_n}} \left(\int_0^1 dt + \int_1^{T_n} \frac{dt}{t^2}\right) \\ &\le C\frac{\mathfrak{e}_n}{\sqrt{v_n}} = o(L^{-1/2})\end{aligned}$$

by (5) and (10). Lemma 4.1 gives

$$\int_{-T_n}^{T_n} \frac{|\psi_n(t) - e^{-t^2/2}|}{|t|}\, dt \le \frac{C}{\sqrt{L}} \int_{\mathbb{R}} t^2 e^{-ct^2}\, dt \le \frac{C}{\sqrt{L}}.$$

Finally, $T_n^{-1} = O(L^{-1/2})$. Theorem 1.1 follows. □

Remark 4.3 (The reset depth is not rate-limiting). The estimate (5) implies

$$\frac{\mathfrak{e}_n}{\sqrt{v_n}} \le C_\kappa e^{-c_\kappa \sqrt{\log L}} L^{-1/2} = o(L^{-1/2}).$$

Thus the reset depth $m \asymp \sqrt{\log(n/a)}$ used in the main paper is already sufficient for the proxy Berry–Esseen theorem. Increasing it to order $\log(n/a)$ may be useful for stronger process approximations, but it is unnecessary for Theorem 1.1.

5 Transfer to the permutation order

We first record a general perturbation inequality.

Lemma 5.1 (Centered L^1 transfer)**.** *Let X and Z be real random variables, and suppose*

$$\mathrm{d_K}(X, N(0,1)) \le \varepsilon, \qquad \mathbb{E}|Z| \le \delta.$$

Then

$$\mathrm{d_K}(X + Z, N(0,1)) \le \varepsilon + C\sqrt{\delta}. \tag{18}$$

Proof. For every $h > 0$ and $x \in \mathbb{R}$,

$$\begin{aligned}\mathbb{P}(X + Z \le x) &\le \mathbb{P}(X \le x + h) + \mathbb{P}(|Z| > h), \\ \mathbb{P}(X + Z \le x) &\ge \mathbb{P}(X \le x - h) - \mathbb{P}(|Z| > h).\end{aligned}$$

The normal distribution function is $1/\sqrt{2\pi}$-Lipschitz, so

$$\mathrm{d_K}(X + Z, N(0,1)) \le \varepsilon + Ch + \frac{\delta}{h}.$$

Choose $h = \sqrt{\delta}$ when $\delta \le 1$; the remaining case is absorbed by the constant. □

Recall that

$$R_n = \log O_n - B_n^\circ, \qquad m_n = \mathbb{E}\log O_n.$$

By Input Theorem 2.2,

$$\mathbb{E}|R_n| \le C_\kappa L \log L. \tag{19}$$

At $s = 0$, the ODE (3) gives

$$|\mathbb{E}B_n^\circ - \mu_n| = |E_n(0)| \le \mathfrak{e}_n. \tag{20}$$

Define

$$X_n = \frac{B_n^\circ - \mu_n}{\sqrt{v_n}}$$

and

$$Z_n^* = \frac{R_n - \mathbb{E}R_n + \mu_n - \mathbb{E}B_n^\circ}{\sqrt{v_n}}. \tag{21}$$

Then exactly

$$\frac{\log O_n - m_n}{\sqrt{v_n}} = X_n + Z_n^*. \tag{22}$$

Moreover, by (19), (20), and (10),

$$\mathbb{E}|Z_n^*| \le \frac{2\mathbb{E}|R_n| + \mathfrak{e}_n}{\sqrt{v_n}} \le C_\kappa \frac{\log L}{\sqrt{L}}. \tag{23}$$

Proof of Theorem 1.2. Apply Lemma 5.1 to (22), use Theorem 1.1, and then (23). This gives

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{v_n}}, N(0,1)\right) \le C_\kappa \left(L^{-1/2} + \left(\frac{\log L}{\sqrt{L}}\right)^{1/2}\right),$$

which is the asserted bound. □

Corollary 5.2 (Classical centering and scale)**.** *One has*

$$\mathrm{d_K}\left(\frac{\log O_n - \frac{1}{2}L^2}{\sqrt{L^3/3}}, N(0,1)\right) \le C_\kappa L^{-1/4}\sqrt{\log L}.$$

Proof. Equations (9), (19), and (20) imply

$$|m_n - \tfrac{1}{2}L^2| \le C_\kappa L \log L.$$

Also, (10) gives

$$\frac{\sqrt{v_n}}{\sqrt{L^3/3}} = 1 + O\left(\frac{\log L}{L}\right).$$

If a stays in a fixed compact subset of $(0,\infty)$, then

$$\sup_x |\Phi((x-b)/a) - \Phi(x)| \le C(|a-1| + |b|),$$

where Φ is the standard normal distribution function. Applying this normal shift-and-scale bound to Theorem 1.2, the additional error is

$$O_\kappa\left(\frac{\log L}{\sqrt{L}}\right),$$

which is smaller than $L^{-1/4}\sqrt{\log L}$. □

6 Moment transfer and the fixed-moment hierarchy

We first retain the general transfer principle from the first checkpoint.

Lemma 6.1 (L^k transfer)**.** *Let $k \geq 1$. If*

$$\mathrm{d_K}(X, N(0,1)) \leq \varepsilon, \qquad \|Z\|_k \leq d,$$

then

$$\mathrm{d_K}(X + Z, N(0,1)) \leq \varepsilon + C_k d^{k/(k+1)}. \tag{24}$$

Proof. For every $h > 0$, the proof of Lemma 5.1 gives

$$\mathrm{d_K}(X + Z, N(0,1)) \leq \varepsilon + Ch + \mathbb{P}(|Z| > h) \leq \varepsilon + Ch + \frac{d^k}{h^k}.$$

Choose $h = d^{k/(k+1)}$. □

Criterion 6.2 (Moment-to-rate criterion)**.** *Suppose that, for a fixed $k \geq 1$ and $d \geq 0$,*

$$\|R_n - \mathbb{E}R_n\|_k \leq C_{\kappa,k} L (\log L)^d. \tag{25}$$

Then

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{v_n}}, N(0,1)\right) \leq C_{\kappa,k}\left[L^{-1/2} + L^{-k/(2(k+1))}(\log L)^{dk/(k+1)}\right]. \tag{26}$$

Proof. By (22), the perturbation of the proxy is

$$Z_n^* = \frac{R_n - \mathbb{E}R_n + \mu_n - \mathbb{E}B_n^\circ}{\sqrt{v_n}}.$$

The deterministic mean mismatch satisfies $|\mu_n - \mathbb{E}B_n^\circ| \leq \mathfrak{e}_n = o(L)$, while $\sqrt{v_n} \asymp L^{3/2}$. Divide (25) by $\sqrt{v_n}$ and apply Lemma 6.1 together with Theorem 1.1. □

The next section supplies factorial domination for every fixed number of cycles. Section 8 gives the transparent $k = 2$ case, and Section 9 proves (25) with $d = 1$ for every fixed integer $k \geq 1$.

7 A fixed-order multi-clock switching theorem

The two-clock one-cycle identity and the three-clock identity of the companion paper are the first cases of a general finite-order construction. The four-cycle case gives the variance checkpoint, while the full fixed-q statement is the input needed for the all-moment argument.

For a coordinate permutation $\tau \in S_n$, define

$$(T_\tau x)_a = x_{\tau(a)}$$

and, if p_n is the joint density of the independent clocks, put

$$R_\tau(x) = \frac{p_n(T_\tau x)}{p_n(x)} = \exp\left\{\sum_{a=1}^{n} \theta_{n,a}(x_a - x_{\tau(a)})\right\}. \tag{27}$$

Then

$$\mathbb{E}\big[R_\tau(X) F(T_\tau X)\big] = \mathbb{E}F(X) \tag{28}$$

for every nonnegative measurable F.

Give the labels a deterministic total order by sorting first by decreasing rate and then by increasing label. For distinct labels $z_0, \dots, z_q$, let

$$h_0, h_1, \dots, h_q$$

be those labels in this order and define the rate-oriented cycle

$$\tau^\star(z_0, \dots, z_q) = (h_0\, h_1 \cdots h_q). \tag{29}$$

Cycle notation is understood up to cyclic rotation.

For positive integers $r_1, \dots, r_q$, let

$$\mathcal{C}_n(r_1, \dots, r_q) = \sum_{\substack{(C_1,\dots,C_q)\in \mathrm{Cyc}(\sigma_n)^q \\ C_1,\dots,C_q \text{ distinct}}} \prod_{j=1}^{q} \mathbf{1}_{\{|C_j|=r_j\}}. \tag{30}$$

This is an ordered distinct-cycle count, even when some r_j coincide.

Theorem 7.1 (Exact fixed-order multi-clock identity)**.** *Fix $q \geq 1$ and positive integers $r_1, \dots, r_q$. Put*

$$R = r_1 + \cdots + r_q.$$

There is a nonnegative random sum $\mathcal{I}^{(q)}_{n;\mathbf{r}}$ with at most $q!n$ summands such that, whenever $R < n$,

$$\left(\prod_{j=1}^{q} r_j\right)(n-R)\mathbb{E}\mathcal{C}_n(r_1, \dots, r_q) = \mathbb{E}\mathcal{I}^{(q)}_{n;\mathbf{r}}. \tag{31}$$

Moreover, for $\kappa > 1$ and $u \geq 0$,

$$\frac{1}{n}\mathbb{E} \sum_{\text{summands in } \mathcal{I}^{(q)}_{n;\mathbf{r}}} \mathbf{1}_{\{\log R_\tau > u\}} \leq q!(q+1)e^{-\lambda_\kappa u}, \tag{32}$$

where $\lambda_\kappa = \kappa/(\kappa-1)$. Consequently,

$$\mathbb{E}\mathcal{I}^{(q)}_{n;\mathbf{r}} \leq C_{\kappa,q} n. \tag{33}$$

For $\kappa = 1$, every factor is one and (33) holds as well.

Proof. Let $\pi \in S_q$. For a source label i with $L_n(i) > R$, put

$$s_{\pi,j} = \sum_{\ell=1}^{j} r_{\pi(\ell)}, \qquad z_{\pi(j)} = \sigma_n^{s_{\pi,j}}(i), \qquad 1 \leq j \leq q,$$

and define

$$\tau_{i,\pi} = (i\, z_{\pi(q)}\, z_{\pi(q-1)} \cdots z_{\pi(1)}). \tag{34}$$

Set

$$\mathcal{I}^{(q)}_{n;\mathbf{r}} = \sum_{i:L_n(i)>R} \sum_{\pi\in S_q} \mathbf{1}_{\{\tau^\star(i,z_1,\dots,z_q)=\tau_{i,\pi}\}} R_{\tau_{i,\pi}}(X). \tag{35}$$

There are at most $q!n$ summands.

For a target permutation, mark an ordered q-tuple of distinct cycles of lengths $r_1, \dots, r_q$, choose a marked label z_j in the jth cycle, and choose i outside their union. The number of such target marks is

$$(n - R) \left(\prod_{j=1}^{q} r_j \right) \mathcal{C}_n(r_1, \dots, r_q).$$

For each deterministic target mark, rotate $\tau^\star(i, z_1, \dots, z_q)$ so that it begins at i. There is a unique $\pi \in S_q$ for which

$$\tau^\star = (i\, z_{\pi(q)} \cdots z_{\pi(1)}).$$

Right composition by $\tau^\star$ cuts a single source cycle whose marked points occur in the order

$$i, z_{\pi(1)}, \dots, z_{\pi(q)}$$

and whose successive segment lengths are $r_{\pi(1)}, \dots, r_{\pi(q)}$ and the residual length. Indeed, $\tau^\star$ replaces the outgoing image of i by the old outgoing image of $z_{\pi(q)}$ and the outgoing image of $z_{\pi(j)}$ by that of $z_{\pi(j-1)}$. The resulting cycles are exactly the q marked segments and the residual segment. Conversely, right composition by $(\tau^\star)^{-1}$ merges those $q+1$ target cycles uniquely. Thus the marked transformation is bijective. Summing (28) over target marks and reorganizing by the source gives (31).

For the tail bound, write the rate-oriented cycle as $\tau = (h_0\, h_1 \cdots h_q)$ with

$$\theta_{h_0} \geq \theta_{h_1} \geq \cdots \geq \theta_{h_q}.$$

Then

$$\begin{aligned} \log R_\tau &= \sum_{j=0}^{q} \theta_{h_j} (X_{h_j} - X_{h_{j+1}}) \\ &= (\theta_{h_0} - \theta_{h_q}) X_{h_0} + \sum_{j=1}^{q} (\theta_{h_j} - \theta_{h_{j-1}}) X_{h_j}, \qquad h_{q+1} = h_0. \end{aligned}$$

Only the coefficient of X_{h_0} is positive. If it is zero, then $R_\tau \leq 1$. Otherwise,

$$\{\log R_\tau > u\} \subseteq \left\{ X_{h_0} > \frac{u}{\theta_{h_0} - \theta_{h_q}} \right\} \subseteq \left\{ X_{h_0} > \frac{u}{\theta_{h_0} - 1} \right\},$$

whose probability is at most $e^{-\lambda_\kappa u}$.

Partition the selected summands by $\pi \in S_q$ and by the position of h_0 among $i, z_1, \dots, z_q$. Within one class, $h_0 = \sigma_n^t(i)$ for one fixed $t \in \{0, s_{\pi,1}, \dots, s_{\pi,q}\}$. Since σ_n^t is a permutation, its restriction to the selected source set is injective. Reindexing by h_0 gives at most $ne^{-\lambda_\kappa u}$ per class, and there are $q!(q+1)$ classes. This proves (32).

Finally, for each nonnegative factor Z,

$$Z \leq 1 + \int_0^\infty e^u \mathbf{1}_{\{\log Z > u\}}\, du.$$

Use (32) and $\lambda_\kappa > 1$ to obtain (33). □

Corollary 7.2 (Fixed-order factorial domination)**.** *For every fixed $q \geq 1$, there is $C_{\kappa,q} < \infty$ such that*

$$\mathbb{E}\mathcal{C}_n(r_1, \dots, r_q) \leq \frac{C_{\kappa,q}}{r_1 \cdots r_q} \tag{36}$$

whenever $r_1 + \cdots + r_q \leq n/2$. Consequently, if $f_1, \dots, f_q$ are nonnegative functions supported on $\{1, \dots, W\}$ and $qW \leq n/2$, then

$$\mathbb{E} \sum_{\substack{(C_1,\dots,C_q)\in \mathrm{Cyc}(\sigma_n)^q \\ C_1,\dots,C_q \text{ distinct}}} \prod_{j=1}^{q} f_j(|C_j|) \leq C_{\kappa,q} \prod_{j=1}^{q} \left(\sum_{r \leq W} \frac{f_j(r)}{r} \right). \tag{37}$$

Proof. Equation (36) follows from Theorem 7.1 and $n/(n - r_1 - \cdots - r_q) \leq 2$. Summing it against $\prod_j f_j(r_j)$ gives (37). □

8 Second moments of the omitted-cycle and arithmetic remainders

We now prove the variance estimate needed for the next checkpoint. Recall

$$\Delta_n = B_n - B_n^\circ, \qquad G_n = B_n^\circ - \log O_n^\circ,$$

and $|R_n| \leq \Delta_n + G_n$.

Lemma 8.1 (Second moment of the omitted-cycle term)**.** *One has*

$$\mathbb{E}\Delta_n^2 \leq C_\kappa L^2 (\log L)^2. \tag{38}$$

Proof. Write

$$\Delta_n = \Delta_n^- + \Delta_n^+ + V_n,$$

where the three sums range over $r \leq b_n$, $U_n < r \leq n/4$, and $r > n/4$, respectively. Deterministically, $V_n \leq 4L$.

For a nonnegative weight f supported on lengths at most $n/4$, Corollary 7.2 with $q = 2$, together with the one-cycle bound $\mathbb{E}C_{n,r} \leq C_\kappa/r$, gives

$$\mathbb{E}\left(\sum_r f(r) C_{n,r} \right)^2 \leq C_\kappa \left(\sum_r \frac{f(r)}{r} \right)^2 + C_\kappa \sum_r \frac{f(r)^2}{r}. \tag{39}$$

For $f(r) = \log r \, \mathbf{1}_{\{r \leq b_n\}}$, the right side is $O((\log L)^4)$. For $f(r) = \log r \, \mathbf{1}_{\{U_n < r \leq n/4\}}$, it is $O(L^2 (\log L)^2)$, because

$$\sum_{U_n < r \leq n/4} \frac{\log r}{r} = O(L \log L),$$
$$\sum_{U_n < r \leq n/4} \frac{(\log r)^2}{r} = O(L^2 \log L).$$

The inequality $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$ completes the proof. □

For the arithmetic term, let

$$\mathcal{R}_n = \{r : b_n < r \le U_n\}, \qquad D_{n,m} = \sum_{\substack{r \in \mathcal{R}_n \\ m \mid r}} C_{n,r}.$$

Then

$$G_n = \sum_{m \le U_n} \Lambda(m)(D_{n,m} - 1)_+, \tag{40}$$

where Λ is the von Mangoldt function.

Lemma 8.2 (Prime-power least-common-multiple sums)**.** *Let $y \ge 2$ and let $\mathcal{Q}_y$ be the set of prime powers larger than y. Then*

$$\sum_{m,m' \in \mathcal{Q}_y} \frac{\Lambda(m)\Lambda(m')}{\operatorname{lcm}(m,m')^2} \le C\frac{\log y}{y}, \tag{41}$$

$$\sum_{m,m' \in \mathcal{Q}_y} \frac{\Lambda(m)\Lambda(m')}{mm' \operatorname{lcm}(m,m')} \le C\frac{\log y}{y^2}. \tag{42}$$

Proof. The Chebyshev estimate $\sum_{k \le x} \Lambda(k) = O(x)$ and partial summation give

$$\sum_{m > y} \frac{\Lambda(m)}{m^2} = O(1/y). \tag{43}$$

If m and m' are powers of different primes, then $\operatorname{lcm}(m,m') = mm'$, and their contribution to either sum is $O(y^{-2})$ by (43).

It remains to take $m = p^a$, $m' = p^b$. By symmetry assume $a \le b$. For (41), the same-prime contribution is at most

$$2\sum_p (\log p)^2 \sum_{b : p^b > y} b p^{-2b}.$$

The terms with $b = 1$ satisfy

$$\sum_{p > y} \frac{(\log p)^2}{p^2} = O\left(\frac{\log y}{y}\right)$$

by partial summation from $\sum_{p \le x} \log p = O(x)$. For $b \ge 2$, use $p^{-2b} \le y^{-1} p^{-b}$ and

$$\sum_p \sum_{b \ge 2} b (\log p)^2 p^{-b} < \infty.$$

This proves (41).

For (42), the same-prime contribution is at most

$$2\sum_p (\log p)^2 \sum_{b : p^b > y} p^{-2b} \sum_{\substack{a \le b \\ p^a > y}} p^{-a} \le \frac{C}{y} \sum_p (\log p)^2 \sum_{b : p^b > y} p^{-2b}$$
$$\le C\frac{\log y}{y^2},$$

using the estimate just proved. This completes the proof. □

For $m \le U_n$, define

$$\alpha_m = \sum_{\substack{r \in \mathcal{R}_n \\ m \mid r}} \frac{1}{r}. \tag{44}$$

Then

$$\alpha_m \le C \frac{\log(1 + U_n/m)}{m} \le C \frac{L}{m}. \tag{45}$$

Lemma 8.3 (Second moment of the arithmetic overlap)**.** *One has*

$$\mathbb{E} G_n^2 \le C_\kappa L^2 (\log L)^2. \tag{46}$$

Proof. Put $y = L^2$ and split $G_n = G_n^{\le y} + G_n^{> y}$.

For the small prime powers, use

$$G_n^{\le y} \le A_n := \sum_{m \le y} \Lambda(m) D_{n,m} = \sum_{\substack{C \in \mathrm{Cyc}(\sigma_n) \\ |C| \in \mathcal{R}_n}} g_y(|C|),$$

where only cycles with lengths in $\mathcal{R}_n$ are included and

$$g_y(r) = \sum_{\substack{m \le y \\ m \mid r}} \Lambda(m) \le \log r \le L.$$

By (39),

$$\mathbb{E} A_n^2 \le C_\kappa \left(\sum_{r \in \mathcal{R}_n} \frac{g_y(r)}{r} \right)^2 + C_\kappa \sum_{r \in \mathcal{R}_n} \frac{g_y(r)^2}{r}.$$

Now

$$\sum_{r \in \mathcal{R}_n} \frac{g_y(r)}{r} \le L \sum_{m \le y} \frac{\Lambda(m)}{m} \le CL \log y,$$

$$\sum_{r \in \mathcal{R}_n} \frac{g_y(r)^2}{r} \le L \sum_{r \in \mathcal{R}_n} \frac{g_y(r)}{r} \le CL^2 \log y.$$

Thus

$$\mathbb{E} (G_n^{\le y})^2 \le C_\kappa L^2 (\log L)^2. \tag{47}$$

For the large prime powers,

$$(D-1)_+ \le \frac{1}{2} (D)_2,$$

so

$$G_n^{> y} \le \mathcal{H}_n := \frac{1}{2} \sum_{m > y} \Lambda(m) (D_{n,m})_2. \tag{48}$$

For prime powers $m, m' > y$, put $\ell = \mathrm{lcm}(m, m')$. Expanding the two ordered pairs of distinct cycles and classifying by whether their union has four, three, or two cycles, Corollary 7.2 with $q = 4, 3, 2$ gives

$$\mathbb{E}[(D_{n,m})_2 (D_{n,m'})_2] \le C_\kappa \left(\alpha_m^2 \alpha_{m'}^2 + \alpha_\ell \alpha_m \alpha_{m'} + \alpha_\ell^2 \right). \tag{49}$$

Indeed, the three-cycle case has four possible cross-identifications and the two-cycle case has two; these fixed multiplicities are absorbed into C_κ.

Using (45), (43), and Lemma 8.2, we obtain

$$\begin{aligned}\mathbb{E}\mathcal{H}_n^2 &\le C_\kappa \left(\sum_{m>y} \Lambda(m)\alpha_m^2\right)^2 \\ &\quad + C_\kappa L^3 \sum_{m,m'>y} \frac{\Lambda(m)\Lambda(m')}{mm' \operatorname{lcm}(m,m')} \\ &\quad + C_\kappa L^2 \sum_{m,m'>y} \frac{\Lambda(m)\Lambda(m')}{\operatorname{lcm}(m,m')^2} \\ &\le C_\kappa \left(\frac{L^4}{y^2} + \frac{L^3 \log y}{y^2} + \frac{L^2 \log y}{y}\right) \\ &\le C_\kappa (1 + \log L).\end{aligned}$$

Together with (47) and $(x+y)^2 \le 2x^2 + 2y^2$, this proves (46). □

Theorem 8.4 (Variance of the centered arithmetic remainder)**.** *One has*

$$\operatorname{Var}(R_n) \le C_\kappa L^2 (\log L)^2. \tag{50}$$

Proof. By (7), $|R_n| \le \Delta_n + G_n$. Hence

$$\operatorname{Var}(R_n) \le \mathbb{E}R_n^2 \le 2\mathbb{E}\Delta_n^2 + 2\mathbb{E}G_n^2.$$

Apply Lemmas 8.1 and 8.3. □

Proof of Theorem 1.3. By Theorem 8.4,

$$\|R_n - \mathbb{E}R_n\|_2 \le C_\kappa L \log L.$$

Apply Criterion 6.2 with $k = 2$ and $d = 1$ to obtain

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{v_n}}, N(0,1)\right) \le C_\kappa \left[L^{-1/2} + L^{-1/3}(\log L)^{2/3}\right].$$

The second term dominates. The classical centering and scaling follow as in Corollary 5.2, because the resulting deterministic shift-and-scale error is $O_\kappa(L^{-1/2} \log L)$. □

9 All fixed moments of the arithmetic remainder

Throughout this section, the moment order k is fixed as $n \to \infty$; constants may depend on k. In particular, for all sufficiently large n, the support conditions $kb_n \le n/2$ and $2kU_n \le n/2$ hold.

9.1 Set-partition bounds for linear cycle statistics

Let $\mathfrak{P}_k$ denote the set of set partitions of $\{1,\dots,k\}$.

Lemma 9.1 (Set-partition moment bound)**.** *Let f be a nonnegative function supported on $\{1,\dots,W\}$, where $kW \le n/2$, and put*

$$S_f = \sum_{C\in \mathrm{Cyc}(\sigma_n)} f(|C|), \qquad A_j(f) = \sum_{r\le W} \frac{f(r)^j}{r}.$$

Then

$$\mathbb{E}S_f^k \le C_{\kappa,k} \sum_{\pi\in\mathfrak{P}_k} \prod_{B\in\pi} A_{|B|}(f). \tag{51}$$

Proof. Expand S_f^k as a sum over ordered k-tuples of cycles and partition the slots according to which slots contain the same cycle. For a fixed $\pi \in \mathfrak{P}_k$, order its blocks by their least elements. The corresponding term is a sum over $|\pi|$ distinct cycles, with the weight $f^{|B|}$ attached to the cycle indexed by the block B. Since $|\pi|W \le kW \le n/2$, Corollary 7.2 bounds that term by the corresponding product in (51). Summing over π proves the result. □

9.2 All moments of the omitted-cycle term

Lemma 9.2 (All fixed moments of the omitted-cycle term)**.** *For every fixed integer $k \ge 1$,*

$$\mathbb{E}\Delta_n^k \le C_{\kappa,k} L^k (\log L)^k. \tag{52}$$

Proof. Put $W_{n,k} = \lfloor n/(2k)\rfloor$ and write

$$\Delta_n = \Delta_{n,k}^- + \Delta_{n,k}^+ + V_{n,k},$$

where the three sums range over $r \le b_n$, $U_n < r \le W_{n,k}$, and $r > W_{n,k}$, respectively. Fewer than $2k$ cycles can have length larger than $W_{n,k}$, so

$$V_{n,k} \le 2kL. \tag{53}$$

For $j \ge 1$, integral comparison gives

$$\sum_{r\le b_n} \frac{(\log r)^j}{r} \le C_j (\log L)^{j+1}, \tag{54}$$

$$\sum_{U_n<r\le W_{n,k}} \frac{(\log r)^j}{r} \le C_{j,k} L^j \log L. \tag{55}$$

For the second estimate, use $\log U_n = L - A_0 \log L + O(1)$ and $\log W_{n,k} = L + O_k(1)$.

Apply Lemma 9.1 first to $f(r) = \log r\, \mathbf{1}_{\{r\le b_n\}}$. A partition π contributes at most

$$C_{\kappa,k} (\log L)^{\sum_{B\in\pi}(|B|+1)} = C_{\kappa,k} (\log L)^{k+|\pi|},$$

so

$$\mathbb{E}(\Delta_{n,k}^-)^k \le C_{\kappa,k} (\log L)^{2k}.$$

Applying the same lemma to $f(r) = \log r\, \mathbf{1}_{\{U_n<r\le W_{n,k}\}}$, a partition π contributes at most

$$C_{\kappa,k} L^k (\log L)^{|\pi|},$$

and hence

$$\mathbb{E}(\Delta_{n,k}^+)^k \le C_{\kappa,k} L^k (\log L)^k.$$

Combine these estimates with (53) and $(x+y+z)^k \le 3^{k-1}(x^k+y^k+z^k)$. □

9.3 Small prime powers

Put $y = L^2$ and split

$$G_n = G_n^{\le y} + G_n^{>y}$$

according to whether the prime power in (40) is at most or larger than y.

Lemma 9.3 (All fixed moments of the small-prime-power overlap)**.** *For every fixed integer $k \ge 1$,*

$$\mathbb{E}(G_n^{\le y})^k \le C_{\kappa,k} L^k (\log L)^k. \tag{56}$$

Proof. As in the second-moment proof,

$$G_n^{\le y} \le A_{n,y} := \sum_{\substack{C \in \mathrm{Cyc}(\sigma_n) \\ |C| \in \mathcal{R}_n}} g_y(|C|), \qquad g_y(r) = \sum_{\substack{m \le y \\ m \mid r}} \Lambda(m).$$

The identity $\sum_{m|r} \Lambda(m) = \log r$ gives $0 \le g_y(r) \le \log r \le L$. Define

$$M_j(y) = \sum_{r \in \mathcal{R}_n} \frac{g_y(r)^j}{r}.$$

Using (45) and the Chebyshev estimate,

$$M_1(y) = \sum_{m \le y} \Lambda(m)\alpha_m \le CL \sum_{m \le y} \frac{\Lambda(m)}{m} \le CL \log y.$$

Therefore, for every $j \ge 1$,

$$M_j(y) \le L^{j-1} M_1(y) \le CL^j \log L. \tag{57}$$

For fixed k, the statistic $A_{n,y}$ is supported on lengths at most U_n, and $kU_n \le n/2$ for all sufficiently large n. Apply Lemma 9.1. By (57), a partition π contributes at most $C_{\kappa,k} L^k (\log L)^{|\pi|}$, which proves (56). □

9.4 Prime-class graph sums

Lemma 9.4 (One-prime graph sum)**.** *Let $H = (V, E)$ be a fixed loopless multigraph with no isolated vertices, $s = |E| \ge 1$, and $q = |V| \ge 2$. For $y \ge 2$, put*

$$\mathcal{Q}_H(y) = \sum_{\mathfrak{p}} (\log \mathfrak{p})^s \sum_{\substack{(a_e)_{e \in E} \in \mathbb{N}^E \\ \mathfrak{p}^{a_e} > y \text{ for all } e}} \mathfrak{p}^{-\sum_{x \in V} \max_{e \ni x} a_e},$$

where $\mathfrak{p}$ ranges over primes. Then

$$\mathcal{Q}_H(y) \le C_H \frac{(\log y)^{s-1}}{y^{q-1}}. \tag{58}$$

Proof. For a prime $\mathfrak{p}$, let

$$h_{\mathfrak{p}} = \min\{a \ge 1 : \mathfrak{p}^a > y\}.$$

If $j = \max_{e \in E}(a_e - h_{\mathfrak{p}})$, then every vertex contributes the baseline $h_{\mathfrak{p}}$, and the two endpoints of an edge attaining the maximum each contribute an additional j. Hence

$$\sum_{x \in V} \max_{e \ni x} a_e \ge q h_{\mathfrak{p}} + 2j.$$

There are at most $(j+1)^s$ exponent vectors with this value of the maximum. Consequently,

$$\sum_{\substack{(a_e)_{e\in E}\in\mathbb{N}^E\\ \mathfrak{p}^{a_e}>y \text{ for all } e}} \mathfrak{p}^{-\sum_x \max_{e\ni x} a_e} \le C_s \mathfrak{p}^{-qh_{\mathfrak{p}}}. \tag{59}$$

If $\mathfrak{p} > y$, then $h_{\mathfrak{p}} = 1$, and partial summation from $\sum_{\mathfrak{p}\le x} \log \mathfrak{p} = O(x)$ gives

$$\sum_{\mathfrak{p}>y} \frac{(\log \mathfrak{p})^s}{\mathfrak{p}^q} \le C_{s,q} \frac{(\log y)^{s-1}}{y^{q-1}}.$$

If $\mathfrak{p} \le y$, then

$$\mathfrak{p}^{-qh_{\mathfrak{p}}} \le y^{-(q-1)} \mathfrak{p}^{-h_{\mathfrak{p}}}$$

and $\mathfrak{p}^{-h_{\mathfrak{p}}} < y^{-1}$. Thus

$$\sum_{\mathfrak{p}\le y} (\log \mathfrak{p})^s \mathfrak{p}^{-qh_{\mathfrak{p}}} \le \frac{(\log y)^{s-1}}{y^q} \sum_{\mathfrak{p}\le y} \log \mathfrak{p} \le C \frac{(\log y)^{s-1}}{y^{q-1}}.$$

Combine these bounds with (59). □

9.5 All moments of the large-prime-power overlap

Lemma 9.5 (All fixed moments of the large-prime-power overlap)**.** *For every fixed integer $k \ge 1$,*

$$\mathbb{E}(G_n^{>y})^k \le C_{\kappa,k}(1+\log L)^{k-1}. \tag{60}$$

Proof. By (48),

$$G_n^{>y} \le \mathcal{H}_n = \frac{1}{2} \sum_{y<m\le U_n} \Lambda(m)(D_{n,m})_2.$$

Expand $\mathcal{H}_n^k$. Each factor $(D_{n,m_e})_2$ contributes an ordered pair of distinct cycle slots. An identification pattern among the $2k$ slots determines a loopless multigraph Γ with k labeled edges and v nonisolated vertices, where $2 \le v \le 2k$. Endpoint order contributes only a finite k-dependent multiplicity, so there are finitely many relevant patterns for fixed k.

For a vertex $x \in V(\Gamma)$, put

$$\ell_x = \operatorname{lcm}\{m_e : e \ni x\},$$

and extend α_m by zero for $m > U_n$. Applying Corollary 7.2 to the distinct cycles represented by the vertices gives

$$\mathbb{E}\prod_{e=1}^{k}(D_{n,m_e})_2 \le C_{\kappa,k} \sum_{\Gamma} \prod_{x\in V(\Gamma)} \alpha_{\ell_x} \le C_{\kappa,k} \sum_{\Gamma} L^v \prod_{x\in V(\Gamma)} \ell_x^{-1}. \tag{61}$$

Here $vU_n \le 2kU_n \le n/2$ for all sufficiently large n.

Fix one identification graph Γ and partition its edge set according to the base primes of the prime powers m_e. Let the resulting nonempty classes be $B_1, \dots, B_c$. For class B_i, let H_i be the loopless subgraph consisting of those edges and their incident vertices, and put

$$s_i = |B_i|, \qquad q_i = |V(H_i)|.$$

Distinct prime bases factor multiplicatively in $\prod_x \ell_x$. On that factorized sum, drop both the upper bounds $m_e \le U_n$ and the requirement that the primes assigned to different classes be distinct. Applying Lemma 9.4 to each class then gives

$$\sum_{\substack{m_1,\dots,m_k>y \\ \text{with the fixed base-prime partition}}} \frac{\prod_{e=1}^{k}\Lambda(m_e)}{\prod_{x\in V(\Gamma)}\ell_x} \le C_k \prod_{i=1}^{c} \frac{(\log y)^{s_i-1}}{y^{q_i-1}} = C_k \frac{(\log y)^{k-c}}{y^S}, \tag{62}$$

where

$$S = \sum_{i=1}^{c}(q_i - 1).$$

Every class contains an edge, so $q_i \ge 2$ and hence $S \ge c$. Every vertex of Γ occurs in at least one class, so $\sum_i q_i \ge v$ and $S \ge v - c$. Therefore

$$S \ge \max\{c, v-c\} \ge \frac{v}{2}.$$

Since $y = L^2$,

$$L^v y^{-S} = L^{v-2S} \le 1.$$

Equations (61) and (62), followed by summation over the finitely many identification patterns and edge partitions, yield

$$\mathbb{E}\mathcal{H}_n^k \le C_{\kappa,k}(1+\log L)^{k-1}.$$

This proves (60). □

9.6 Completion of the fixed-moment hierarchy

Proof of Theorem 1.4. By Lemmas 9.3 and 9.5,

$$\mathbb{E}G_n^k \le C_{\kappa,k} L^k (\log L)^k.$$

Together with Lemma 9.2 and the deterministic bound $|R_n| \le \Delta_n + G_n$, this gives

$$\mathbb{E}|R_n|^k \le C_{\kappa,k} L^k (\log L)^k.$$

Finally,

$$\|R_n - \mathbb{E}R_n\|_k \le \|R_n\|_k + |\mathbb{E}R_n| \le 2\|R_n\|_k,$$

which proves the centered estimate. □

Proof of Theorem 1.5. Apply Criterion 6.2 with $d = 1$ and the moment order k. For the classical centering and scale, use

$$|m_n - \tfrac{1}{2}L^2| \le C_\kappa L \log L, \qquad \frac{\sqrt{v_n}}{\sqrt{L^3/3}} = 1 + O\left(\frac{\log L}{L}\right).$$

The resulting normal shift-and-scale error is

$$O_\kappa(L^{-1/2}\log L).$$

For every fixed k, this is

$$o\left(L^{-k/(2(k+1))}(\log L)^{k/(k+1)}\right).$$

□

Corollary 9.6 (Third-moment checkpoint)**.** *One has*

$$\mathrm{d_K}\left(\frac{\log O_n - m_n}{\sqrt{v_n}}, N(0,1)\right) \le C_\kappa L^{-3/8}(\log L)^{3/4},$$

and the same rate holds with the classical centering and scale.

Proof. Take $k = 3$ in Theorem 1.5. □

Proof of Corollary 1.6. Fix $\varepsilon > 0$ and choose an integer k so large that $1/(2(k+1)) < \varepsilon/2$. Then

$$L^{-k/(2(k+1))}(\log L)^{k/(k+1)} = L^{-1/2+1/(2(k+1))}(\log L)^{k/(k+1)} \le L^{-1/2+\varepsilon}$$

for all sufficiently large n. Apply Theorem 1.5; its classical-centering statement gives the second assertion. □

Part I
Sharp rate and first Edgeworth expansion

10 Audit gate I: free-depth endpoint input and polynomial switching

The polynomial approximation requires a stronger specialization of the finite-endpoint construction than the square-root reset used in Theorem 1.1. The source theorem is formulated with a free terminal depth. We record the precise quantitative consequence used here.

For a window $r_- \le r \le r_+$ and a fixed number q of independent roots, put

$$K = qr_+, \qquad \delta = K/n, \qquad h = \delta^{1/4}, \qquad \rho = \delta^{1/(4\kappa)}, \qquad M = \sqrt{nK}.$$

Input theorem 10.1 (Free-depth finite-endpoint comparison)**.** *Let m satisfy*

$$1 \le m \le r_-/3$$

and

$$m \to \infty, \quad mh \to 0, \quad m\rho \to 0, \quad m^2/M \to 0, \quad mL/M \to 0, \quad m/(hM) \to 0.$$

For every fixed q, the conditional endpoint comparison of [7] has error at most

$$\mathcal{E}_{n,q}(m) \le C_{\kappa,q}\Big[(1-\alpha)^{m-1} + mh + \frac{m^2}{M} + m\rho + \frac{mL}{M} + \rho + \delta + \delta^{1/2}\log(1/\delta) + h^{-1}\log(1/\delta)e^{-cM} + n^2h^{-1}e^{-chM} + n^{-2}\Big]. \tag{63}$$

The bound is uniform in every previously exposed compatible terminal history. It also applies to products of bounded Lipschitz functions of the selected swap factors: if $F_A(i,j) = R_{ij} \wedge A$, then the one-root error is at most $C_\kappa A\mathcal{E}_{n,1}(m)$ and the two-root error is at most $C_\kappa A^2\mathcal{E}_{n,2}(m)$.

For every fixed q, Q, on a window

$$L^3 \le r_- \le r_+ \le n/L^A,$$

the choice $m = \lfloor \log(1/\delta) \rfloor$ is admissible and, after choosing $A = A(\kappa, q, Q)$ sufficiently large,

$$\mathcal{E}_{n,q}(m) \le C_{\kappa,q,Q}L^{-Q}.$$

Verification of the polynomial specialization. The complete proof of the free-depth theorem is in [7]; here we verify the specialization and the observable constants used below. On the stated window, $m \le L + O_q(1) \ll r_-$, while

$$M \ge c_q \sqrt{n}\, L^{3/2}, \qquad hM = n\delta^{3/4} \ge c_q n^{1/4} L^{9/4}.$$

Hence every admissibility ratio tends to zero uniformly. Moreover,

$$(1-\alpha)^m \le C\delta^c, \qquad mh \le C(1+\log(1/\delta))\delta^{1/4}, \qquad m\rho \le C(1+\log(1/\delta))\delta^{1/(4\kappa)}.$$

The inverse-row terms are $O(L^{1/2}n^{-1/2})$, and the environment terms are smaller than every fixed power of L^{-1}. Since $\delta \le qL^{-A}$, increasing A makes (63) $O(L^{-Q})$.

For a bounded function of one exact swap factor, replacing an exact clock by its block representative changes $\log R_{ij}$ by $O_\kappa(h)$; the map $z \mapsto e^z \wedge A$ is bounded and A-Lipschitz. Thus the coarse-to-exact replacement costs $O_\kappa(Ah)$, already present in $\mathcal{E}_{n,1}$. Conditional total variation contributes $O(A\mathcal{E}_{n,1})$. A telescoping product gives the corresponding $O(A^2\mathcal{E}_{n,2})$ two-root bound. This proves the stated observable form. □

For a dyadic block $a \le r < 2a$, define

$$X_{n,r}^{(A)} = \frac{1}{n} \sum_{i: L_n(i) > r} F_A(i, \sigma_n^r(i)), \qquad D_{n,A} = \frac{1}{n^2} \sum_{i,j=1}^{n} F_A(i,j), \qquad \mu_{n,A} = \mathbb{E} D_{n,A}.$$

Proposition 10.2 (Polynomial one-block switching)**.** *For every fixed $Q > 0$, after choosing A_0 sufficiently large,*

$$\sup_{b_n \le a \le U_n} \sup_{a \le r < 2a} \mathbb{E} \left| \frac{S_{n,r}}{n} - 1 \right| \le C_{\kappa,Q} L^{-Q-2}.$$

Consequently,

$$\sum_{j: a_j \le U_n} \sup_{a_j \le r < 2a_j} \mathbb{E} \left| \frac{S_{n,r}}{n} - 1 \right| \le C_{\kappa,Q} L^{-Q}, \qquad a_j = 2^j b_n. \tag{64}$$

Proof. Apply Input Theorem 10.1 on $[a, 2a]$, first with one root and then with two independent roots. For any prescribed $M_0 > 0$, increasing A_0 gives

$$|\mathbb{E} X_{n,r}^{(A)} - \mu_{n,A}| \le C_\kappa A L^{-M_0}, \tag{65}$$

$$|\mathbb{E}(X_{n,r}^{(A)})^2 - \mathbb{E} D_{n,A}^2| \le C_\kappa A^2 L^{-M_0}. \tag{66}$$

Replacing one clock changes at most $2n$ summands of $D_{n,A}$, each by at most A, so Efron–Stein gives

$$\mathrm{Var}(D_{n,A}) \le 4A^2/n.$$

Equations (65) and (66) therefore imply

$$\mathbb{E}|X_{n,r}^{(A)} - \mu_{n,A}|^2 \le C_\kappa A^2 (L^{-M_0} + n^{-1}).$$

The selected-pair and fresh-pair tail estimates from the switching manuscript give, for $\kappa > 1$,

$$\mathbb{E} \frac{S_{n,r} - S_{n,r}^{(A)}}{n} + |1 - \mu_{n,A}| \le C_\kappa A^{-1/(\kappa-1)}.$$

Choose $A = L^{(\kappa-1)(Q+4)}$ and then choose M_0 so large that the truncated L^2 term is $O(L^{-Q-4})$. This proves the asserted one-block bound. When $\kappa = 1$, all swap factors equal one and the same conclusion follows with $A = 1$. There are $O(L)$ dyadic blocks, which proves (64). □

11 Audit gate II: polynomial passive-mark product–Poisson approximation

Let

$$\mathcal{I}_n = \{r \in \mathbb{Z} : b_n < r \le U_n\}, \qquad \xi_n = (C_{n,r})_{r \in \mathcal{I}_n}.$$

A permutation-measurable random element $M_n = M_n(\sigma_n)$ is called a *passive mark* if every good split with $r \in \mathcal{I}_n$ and source-cycle length greater than $r + U_n$ leaves M_n unchanged. More general clock-dependent marks would require the marked switching identity to be formulated on the full clock space; no such extension is used here.

Lemma 11.1 (Passive-mark Campbell–Poisson transfer)**.** *Let I be finite, let M be a standard Borel space, let (X, M) take values in $\mathbb{Z}_+^I \times \mathsf{M}$, and let $\lambda_i \ge 0$. Suppose that for every measurable family $g_i : \mathbb{Z}_+^I \times \mathsf{M} \to [-1, 1]$,*

$$\left| \mathbb{E} \sum_{i \in I} X_i g_i(X - e_i, M) - \sum_{i \in I} \lambda_i \mathbb{E} g_i(X, M) \right| \le \varepsilon.$$

If Z_i are independent $\mathrm{Pois}(\lambda_i)$ variables and $\widetilde{M}$ is an independent copy of M, independent of Z, then

$$\mathrm{d_{TV}}\big(\mathcal{L}(X, M), \mathcal{L}(Z, \widetilde{M})\big) \le \varepsilon.$$

The constant is independent of $|I|$ and $\sum_i \lambda_i$.

Proof. Let $A \subseteq \mathbb{Z}_+^I \times \mathsf{M}$ be measurable and, for $m \in \mathsf{M}$, put $f_m(x) = \mathbf{1}_A(x, m)$. Let $\mathcal{A}$ be the immigration–death generator with immigration rates $(\lambda_i)_{i \in I}$, and let P_t be its semigroup. With π denoting the product–Poisson law, define

$$h_A(x, m) = -\int_0^\infty \{P_t f_m(x) - \pi f_m\}\, dt.$$

Coupling the chain started from x to a stationary chain bounds the integrand by $\min\{1, e^{-t}(|x| + \sum_i \lambda_i)\}$, so the integral is finite. Because the first coordinate is countable and M is standard Borel, this is jointly measurable. The usual coupling of chains started from x and $x + e_i$ shows that

$$g_i(x, m) := h_A(x + e_i, m) - h_A(x, m) \quad \text{satisfies} \quad |g_i(x, m)| \le 1$$

uniformly in x, m, i. The Stein equation gives

$$\mathcal{A} h_A(x, m) = f_m(x) - \pi f_m.$$

Consequently,

$$\begin{aligned} &\mathbb{P}\{(X, M) \in A\} - \mathbb{P}\{(Z, \widetilde{M}) \in A\} \\ &\quad = \mathbb{E}\mathcal{A} h_A(X, M) \\ &\quad = \sum_{i \in I} \lambda_i \mathbb{E} g_i(X, M) - \mathbb{E} \sum_{i \in I} X_i g_i(X - e_i, M). \end{aligned}$$

The assumed Campbell discrepancy bounds the absolute value by ε. Taking the supremum over A proves the claim. □

Proof of Theorem 1.7. Let $M_n = M_n(\sigma_n)$ be a permutation-measurable passive mark and let $g_r : \mathbb{Z}_+^{\mathcal{I}_n} \times \mathsf{M} \to [-1, 1]$. Apply the marked one-switch identity with

$$\Phi_r(\pi) = \begin{cases} g_r(\xi(\pi) - e_r, M(\pi)), & C_r(\pi) \geq 1, \\ 0, & C_r(\pi) = 0. \end{cases}$$

On a good split, the core vector increases by e_r and the mark is unchanged. On the exceptional source-cycle range the two test functions have modulus at most one. Exactly as in the unmarked Campbell calculation,

$$\begin{aligned} &\left| \mathbb{E} \sum_{r \in \mathcal{I}_n} C_{n,r} g_r(\xi_n - e_r, M_n) - \sum_{r \in \mathcal{I}_n} a_{n,r} \mathbb{E} g_r(\xi_n, M_n) \right| \\ &\qquad \leq C \sum_j \sup_{a_j \leq r < 2a_j} \mathbb{E} \left| \frac{S_{n,r}}{n} - 1 \right| + C_\kappa L \left(\frac{U_n}{n} \right)^\beta. \end{aligned}$$

The first term is $O(L^{-Q})$ by Proposition 10.2; the second is $L^{1-A_0\beta}$ and is made $O(L^{-Q})$ by increasing A_0. Lemma 11.1 proves the joint product–Poisson estimate.

Taking a constant mark proves the unmarked assertion. For $K_n^+ = \sum_{r > U_n} C_{n,r}$, a good split replaces one upper cycle by one upper residual cycle, so the mark is passive. □

Corollary 11.2 (Polynomial lower tail for the number of core cycles)**.** *Let*

$$K_n^\circ = \sum_{b_n < r \leq U_n} C_{n,r}.$$

Then

$$\mathbb{P}\{K_n^\circ \leq 4\} \leq C_\kappa L^{-3}.$$

Proof. Under the comparison law, the sum of the core coordinates is Poisson with mean

$$\lambda_n = \sum_{b_n < r \leq U_n} a_{n,r} = L - (A_0 + 3) \log L + O(1) \geq L/2$$

for all large n. Its lower tail at four is $O(L^4 e^{-L/2})$. Apply Theorem 1.7 with $Q = 4$ and use contraction of total variation under summation. □

12 Audit gate III: the independent linearized Poisson Edgeworth theorem

Fix once and for all a constant $B_* > 4$, and put

$$y = y_n = \left\lfloor \frac{L}{(\log L)^{B_*}} \right\rfloor.$$

For an integer $r \geq 1$, define

$$g_y(r) = \sum_{\substack{m \leq y \\ m \mid r}} \Lambda(m), \qquad q_y(r) = \log r - g_y(r).$$

The identity $\sum_{m|r}\Lambda(m)=\log r$ gives $0\le q_y(r)\le \log r$.

Let $(N_{n,r})_{r\in\mathcal{I}_n}$ be independent with $N_{n,r}\sim \mathrm{Pois}(a_{n,r})$, and define

$$Y_n=\sum_{r\in\mathcal{I}_n} q_y(r)N_{n,r},\quad \mu_{q,n}=\sum_{r\in\mathcal{I}_n} a_{n,r}q_y(r),\quad \sigma_{q,n}^2=\sum_{r\in\mathcal{I}_n} a_{n,r}q_y(r)^2.$$

Also put

$$\rho_{3,q,n}=\sum_{r\in\mathcal{I}_n} a_{n,r}q_y(r)^3,\qquad \gamma_{q,n}=\rho_{3,q,n}/\sigma_{q,n}^3.$$

Lemma 12.1 (Linearized cumulants)**.** *One has*

$$\sigma_{q,n}^2=\frac{1}{3}L^3+O(L^2\log L), \tag{67}$$

$$\rho_{3,q,n}=\frac{1}{4}L^4+O(L^3\log L), \tag{68}$$

$$\sum_{r\in\mathcal{I}_n} a_{n,r}q_y(r)^4=O(L^5). \tag{69}$$

Consequently,

$$\gamma_{q,n}=\frac{3\sqrt{3}}{4}L^{-1/2}+O\left(\frac{\log L}{L^{3/2}}\right). \tag{70}$$

Proof. For fixed $k\ge 0$, the same integral comparison as in Lemma 3.1 gives

$$\sum_{r\in\mathcal{I}_n} a_{n,r}(\log r)^k=\frac{L^{k+1}}{k+1}+O_k(L^k\log L).$$

For $j\ge 0$,

$$\begin{aligned}\sum_{r\in\mathcal{I}_n} a_{n,r}(\log r)^j g_y(r)&=\sum_{m\le y}\Lambda(m)\sum_{\substack{r\in\mathcal{I}_n\\ m|r}} a_{n,r}(\log r)^j\\ &\le C_jL^{j+1}\sum_{m\le y}\frac{\Lambda(m)}{m}\le C_jL^{j+1}\log y.\end{aligned}$$

Since

$$0\le (\log r)^k-q_y(r)^k\le k(\log r)^{k-1}g_y(r),$$

the assertions for $k=2,3$ follow. The fourth moment follows from $q_y(r)\le\log r$. Dividing (68) by the three-halves power of (67) proves (70). □

Let

$$X_n=\frac{Y_n-\mu_{q,n}}{\sigma_{q,n}},\qquad \phi_n(t)=\mathbb{E}e^{itX_n}.$$

Lemma 12.2 (Low-frequency Edgeworth expansion)**.** *There are $c,C>0$ such that, for $|t|\le c\sqrt{L}$,*

$$|\phi_n(t)|\le e^{-ct^2}.$$

If

$$\widehat{G}_n(t)=e^{-t^2/2}\left(1+\frac{(it)^3}{6}\gamma_{q,n}\right),$$

then

$$\int_{|t|\le c\sqrt{L}}\frac{|\phi_n(t)-\widehat{G}_n(t)|}{|t|}\,dt\le\frac{C}{L}.$$

Proof. The compound-Poisson exponent is

$$\log \phi_n(t) = \sum_{r \in \mathcal{I}_n} a_{n,r} \left(e^{itq_y(r)/\sigma_{q,n}} - 1 - \frac{itq_y(r)}{\sigma_{q,n}} \right).$$

Since $\max_r q_y(r)/\sigma_{q,n} = O(L^{-1/2})$, Taylor expansion on $|t| \le c\sqrt{L}$ gives

$$\log \phi_n(t) = -\frac{1}{2}t^2 + \frac{(it)^3}{6}\gamma_{q,n} + R_n(t), \qquad |R_n(t)| \le Ct^4/L.$$

The cosine bound gives the Gaussian modulus estimate. On $|t| \le L^{1/12}$, exponentiation yields

$$|\phi_n(t) - \widehat{G}_n(t)| \le \frac{C}{L}(t^4 + t^6)e^{-ct^2}.$$

On $L^{1/12} < |t| \le c\sqrt{L}$, both terms have exponentially small modulus. Integration proves the claim. □

Lemma 12.3 (Uniform logarithmic Riemann sum)**.** *Let*

$$\lambda_n = \sum_{r \in \mathcal{I}_n} a_{n,r}, \qquad M_n(u) = \frac{1}{\lambda_n} \sum_{r \in \mathcal{I}_n} a_{n,r} e^{iuq_y(r)/L}.$$

Uniformly for $|u| \le \sqrt{L}/(\log L)^2$,

$$M_n(u) = \int_0^1 e^{iux}\, dx + O\left(\frac{\log L}{L} + \frac{|u| \log y}{L} \right). \tag{71}$$

Proof. Replacing $q_y(r)$ by $\log r$ costs at most

$$\frac{|u|}{L\lambda_n} \sum_{r \in \mathcal{I}_n} a_{n,r} g_y(r) \le C \frac{|u| \log y}{L}.$$

Replacing $a_{n,r}$ by $1/r$ costs $O(U_n/(nL))$. Uniformly on the stated range, the derivative of $x^{-1}e^{iu \log x/L}$ is $O(x^{-2})$, so sum–integral comparison gives

$$\sum_{b_n < r \le U_n} \frac{1}{r} e^{iu \log r/L} = L \int_{\log b_n/L}^{\log U_n/L} e^{iuv}\, dv + O(b_n^{-1}).$$

Now $\lambda_n = L - (A_0 + 3)\log L + O(1)$, while the two omitted endpoint intervals in $[0, 1]$ have total length $O(\log L/L)$. Division by λ_n proves (71). □

Lemma 12.4 (Extended Fourier window)**.** *There are constants* $c_0, c_1, c_2 > 0$ *such that*

$$|\phi_n(t)| \le e^{-c_2 L}$$

whenever

$$c_0\sqrt{L} \le |t| \le T_n, \qquad T_n = \frac{c_1 L}{(\log L)^2}.$$

Proof. Choose $u_0 > 0$. Since

$$\Re \int_0^1 e^{iux}\,dx = \frac{\sin u}{u},$$

there is $\eta > 0$ such that this real part is at most $1-\eta$ for $|u| \geq u_0$. Lemma 12.3 transfers the same gap, with $\eta/2$, uniformly up to $|u| \leq \sqrt{L}/(\log L)^2$. Set $u = tL/\sigma_{q,n}$. The stated t-window maps into this u-window and stays outside $(-u_0, u_0)$ after suitable choices of c_0, c_1. Since

$$\log|\phi_n(t)| = \lambda_n\{\Re M_n(u) - 1\}$$

and $\lambda_n \asymp L$, the conclusion follows. □

Lemma 12.5 (Signed Esseen smoothing)**.** *Let F be a distribution function and let G be the cumulative function of a finite signed measure of total mass one. Assume that G is absolutely continuous with density $g \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$, and assume $\int |x|\,dF(x) + \int |x||g(x)|\,dx < \infty$. Write*

$$\widehat{F}(t) = \int_{\mathbb{R}} e^{itx}\,dF(x), \qquad \widehat{G}(t) = \int_{\mathbb{R}} e^{itx} g(x)\,dx.$$

Then, for every $T > 0$,

$$\sup_x |F(x) - G(x)| \leq C \int_{-T}^{T} \frac{|\widehat{F}(t) - \widehat{G}(t)|}{|t|}\,dt + \frac{C\|g\|_\infty}{T}, \tag{72}$$

where the integrand at zero is interpreted by continuity. The constant is absolute; monotonicity of G is not required.

Proof. Choose an even nonnegative probability density K such that $\widehat{K}$ is supported on $[-1,1]$ and $\int |x|K(x)\,dx < \infty$; a normalized fourth power of a sinc function has these properties. Put $K_T(x) = TK(Tx)$ and $H = F - G$. Since the two underlying signed measures have the same total mass, $H(-\infty) = H(+\infty) = 0$.

We first prove the one-sided smoothing estimate

$$\|H\|_\infty \leq C\|H * K_T\|_\infty + \frac{C\|g\|_\infty}{T}. \tag{73}$$

Let $D = \|H\|_\infty$. If $H(x) \geq D - \varepsilon$, then, for every $z \geq 0$, monotonicity of F and the Lipschitz bound for G give $H(x+z) \geq H(x) - \|g\|_\infty z$. Choose $c > 0$ so that $p := \int_{-\infty}^{c} K(u)\,du > 1/2$, and evaluate the convolution at $x + c/T$. After the change of variables $u = Ty$, the region $u \leq c$ gives the preceding lower bound, while the complementary region is bounded below by $-D$. Hence

$$(H * K_T)(x + c/T) \geq (2p-1)D - p\varepsilon - \frac{\|g\|_\infty}{T} \int_{u \leq c} (c-u)K(u)\,du.$$

Letting $\varepsilon \downarrow 0$ bounds the positive maximum of H. For a point where H is near $-D$, use instead $H(x-z) \leq H(x) + \|g\|_\infty z$ for $z \geq 0$ and evaluate at $x - c/T$. This bounds the negative minimum and proves (73).

Let $\mu = dF - g(x)\,dx$, a finite signed measure of total mass zero. The first-moment assumption implies $H \in L^1(\mathbb{R})$: on each half-line, integrate the corresponding tail bound $|H(x)| \leq |\mu|((x,\infty))$ or $|H(x)| \leq |\mu|((-\infty, x])$. Integration by parts therefore gives

$$\widehat{H}(t) = \frac{\widehat{F}(t) - \widehat{G}(t)}{-it}, \qquad t \neq 0.$$

Since $\widehat{K}(t/T)$ is supported on $[-T,T]$, ordinary Fourier inversion gives

$$(H * K_T)(x) = \frac{1}{2\pi}\int_{-T}^{T} e^{-itx}\frac{\widehat{F}(t)-\widehat{G}(t)}{-it}\widehat{K}(t/T)\,dt.$$

Because $|\widehat{K}| \le 1$, the absolute value is bounded by the integral in (72), up to an absolute constant. Combining this with (73) proves the lemma. This is the standard signed form of Esseen's smoothing argument; compare [8, Chapter VI]. □

Theorem 12.6 (Independent linearized Poisson Edgeworth expansion)**.** *Define*

$$G_n(x) = \Phi(x) + \frac{\gamma_{q,n}}{6}(1-x^2)\varphi(x).$$

Then

$$\sup_x |\mathbb{P}\{X_n \le x\} - G_n(x)| \le C\frac{(\log L)^2}{L}. \tag{74}$$

Consequently, uniformly in x,

$$\mathbb{P}\{X_n \le x\} = \Phi(x) - \frac{\sqrt{3}}{8}(x^2-1)\varphi(x)L^{-1/2} + o(L^{-1/2}). \tag{75}$$

Proof. The signed density of G_n is

$$g_n(x) = G_n'(x) = \varphi(x)\left[1 + \frac{\gamma_{q,n}}{6}(x^3-3x)\right].$$

It belongs to $L^1 \cap L^\infty$, has integral one and a finite first absolute moment, and satisfies $\sup_n \|g_n\|_\infty < \infty$. The compound-Poisson variable X_n also has a finite first absolute moment. Its Fourier transform is $\widehat{G}_n$ from Lemma 12.2. Combine Lemmas 12.2 and 12.4; on the extended portion the Gaussian approximant is also exponentially small. Thus

$$\int_{-T_n}^{T_n} \frac{|\phi_n(t) - \widehat{G}_n(t)|}{|t|}\,dt \le C/L.$$

Lemma 12.5 and $T_n^{-1} = O((\log L)^2/L)$ prove (74). Substitute (70) to obtain (75). □

13 Transfer to the full linearized Luce additive statistic

Let

$$A^\circ_{n,y} = \sum_{r\in\mathcal{I}_n} q_y(r)C_{n,r}, \qquad T_n^\sharp = A^\circ_{n,y} + LK_n^+.$$

Lemma 13.1 (Second moment of the upper-cycle mark)**.** *One has*

$$\mathbb{E}(K_n^+)^2 \le C_\kappa(\log L)^2.$$

Proof. For $U_n < r \le n/4$, the one- and two-cycle factorial estimates give the usual linear-statistic second-moment bound with weight one; the relevant harmonic mass is $O(\log(n/U_n)) = O(\log L)$. There are deterministically at most three cycles longer than $n/4$. Splitting at $n/4$ proves the result. □

Proposition 13.2 (Edgeworth expansion for the marked principal statistic)**.** *Uniformly in x,*

$$\mathbb{P}\left\{\frac{T_n^\sharp - \mathbb{E}T_n^\sharp}{\sigma_{q,n}} \le x\right\} = G_n(x) + o_\kappa(L^{-1/2}). \tag{76}$$

Proof. By Theorem 1.7, the joint law of the core vector and K_n^+ is polynomially close to a product of the Poisson core and an independent copy $\widetilde{K}_n^+$ of the mark. Moreover, the one-switch identity and Proposition 10.2 give

$$|\mathbb{E}A^\circ_{n,y} - \mu_{q,n}| \le C_{\kappa,Q}L^{-Q}.$$

It therefore suffices to study

$$Y_n + L(\widetilde{K}_n^+ - \mathbb{E}K_n^+).$$

Put

$$Z_n = \frac{L(\widetilde{K}_n^+ - \mathbb{E}K_n^+)}{\sigma_{q,n}}.$$

Then $\mathbb{E}Z_n = 0$ and, by Lemma 13.1,

$$\mathbb{E}Z_n^2 \le C_\kappa \frac{(\log L)^2}{L} = o(L^{-1/2}).$$

The functions G_n have uniformly bounded second derivatives, so Taylor's theorem gives

$$\sup_x |\mathbb{E}G_n(x - Z_n) - G_n(x)| \le C\mathbb{E}Z_n^2 = o(L^{-1/2}).$$

Convolving Theorem 12.6 with the independent mark, and then using polynomial total variation and the negligible mean mismatch, proves (76). □

Write

$$A_{n,y} = \sum_{r=1}^{n} q_y(r)C_{n,r} = T_n^\sharp + D_{n,y},$$

where

$$D_{n,y} = \sum_{r\le b_n} q_y(r)C_{n,r} + \sum_{r>U_n} (q_y(r) - L)C_{n,r}.$$

Lemma 13.3 (Omitted linearized additive terms)**.** *Let $W = n/8$. There is a decomposition*

$$D_{n,y} = D^{\mathrm{sm}}_{n,y} + D^{\mathrm{bd}}_{n,y}$$

such that

$$\mathbb{E}|D^{\mathrm{sm}}_{n,y}|^2 \le C_\kappa(\log L)^4, \qquad \|D^{\mathrm{bd}}_{n,y}\|_\infty \le Cy.$$

Proof. Take D^{sm} over $r \le b_n$ and $U_n < r \le W$, and take the remaining $r > W$ contribution as D^{bd}. The short range is bounded by the linear-statistic second-moment inequality with weight $\log r$, giving $O((\log L)^4)$.

For $U_n < r \le W$,

$$|q_y(r) - L| \le L - \log r + g_y(r).$$

Harmonic integration gives

$$\sum_{U_n<r\le W} \frac{L - \log r}{r} = O((\log L)^2), \qquad \sum_{U_n<r\le W} \frac{(L - \log r)^2}{r} = O((\log L)^3).$$

Also

$$\sum_{U_n<r\le W}\frac{g_y(r)}{r}=O((\log L)^2).$$

Expanding $g_y(r)^2$ and summing first over multiples of $\operatorname{lcm}(m,m')$ gives

$$\sum_{U_n<r\le W}\frac{g_y(r)^2}{r}\le C\log L\sum_{m,m'\le y}\frac{\Lambda(m)\Lambda(m')}{\operatorname{lcm}(m,m')}=O((\log L)^3).$$

Here the double prime-power sum is $O((\log y)^2)$: different prime bases factor, and equal prime bases are summed by the larger exponent. Since four lengths in this range have total at most $n/2$, the fixed-order factorial bound gives the stated second moment.

Finally, at most seven cycles have length greater than W, and for such r,

$$|q_y(r)-L|\le\log 8+\psi(y)\le Cy.$$

This proves the bounded part. □

14 Audit gate IV: the centered nonlinear arithmetic remainder

For all cycle lengths define

$$\widehat{D}_{n,m}=\sum_{\substack{1\le r\le n\\ m\mid r}}C_{n,r}.$$

The full product–least-common-multiple identity is

$$\sum_{r=1}^{n}(\log r)C_{n,r}-\log O_n=\sum_{m\le n}\Lambda(m)(\widehat{D}_{n,m}-1)_+.$$

For $m\le y$ use $(D-1)_+=D-1+\mathbf{1}_{\{D=0\}}$. This gives the exact linearization

$$\log O_n=A_{n,y}+\psi(y)-V_{n,y}-H_{n,y},\tag{77}$$

where

$$V_{n,y}=\sum_{m\le y}\Lambda(m)\mathbf{1}_{\{\widehat{D}_{n,m}=0\}},\qquad H_{n,y}=\sum_{m>y}\Lambda(m)(\widehat{D}_{n,m}-1)_+.$$

The vacancy term is deterministically bounded by

$$0\le V_{n,y}\le\psi(y)\le Cy.\tag{78}$$

14.1 The core large-prime-power correction

Let

$$D^{\circ}_{n,m}=\sum_{\substack{r\in\mathcal{I}_n\\ m\mid r}}C_{n,r},\qquad H^{\circ}_{n,y}=\sum_{m>y}\Lambda(m)(D^{\circ}_{n,m}-1)_+.$$

Define $D^{P}_{n,m}$ and $H^{P}_{n,y}$ from the independent Poisson core.

Lemma 14.1 (Product-Poisson Poincaré inequality)**.** *Let N_i be independent Poisson variables with means λ_i. For every square-integrable function F,*

$$\operatorname{Var} F(N) \le \sum_i \lambda_i \mathbb{E}\big|F(N+e_i) - F(N)\big|^2.$$

Proof. By tensorization of conditional variance it suffices to prove the one-dimensional inequality. Let $(P_t)_{0\le t\le\lambda}$ be a unit-rate Poisson process and put $Z = P_\lambda$. For bounded f, the Doob martingale $M_t = \mathbb{E}[f(P_\lambda) \mid \mathcal{F}_t]$ has jump amplitude

$$P_{\lambda-t}(\Delta f)(P_{t-}), \qquad \Delta f(k) = f(k+1) - f(k),$$

where P_s also denotes the Poisson semigroup. The predictable quadratic variation identity and Jensen's inequality give

$$\begin{aligned}\operatorname{Var} f(Z) &= \mathbb{E}\int_0^\lambda \big|P_{\lambda-t}(\Delta f)(P_t)\big|^2\,dt \\ &\le \int_0^\lambda \mathbb{E}P_{\lambda-t}(|\Delta f|^2)(P_t)\,dt = \lambda\mathbb{E}|\Delta f(Z)|^2.\end{aligned}$$

Approximation proves the result for every square-integrable f, and successive conditioning over the independent coordinates gives the stated product inequality. □

Lemma 14.2 (Poisson Poincaré bound for the core correction)**.** *One has*

$$\operatorname{Var}(H^P_{n,y}) \le C\left[L^3\frac{\log y}{y^2} + L^2\frac{\log y}{y}\right] \le CL(\log L)^{2B_*+1}. \tag{79}$$

Proof. Lemma 14.1 gives

$$\operatorname{Var} F(N) \le \sum_r a_{n,r}\mathbb{E}|F(N+e_r) - F(N)|^2.$$

Here adding one r-cycle changes $H^P_{n,y}$ by

$$\sum_{\substack{m>y\\ m\mid r}} \Lambda(m)\mathbf{1}_{\{D^P_{n,m}\ge 1\}}.$$

Put

$$\alpha^P_m = \sum_{\substack{r\in\mathcal{I}_n\\ m\mid r}} a_{n,r} \le CL/m.$$

For $\ell = \operatorname{lcm}(m,m')$,

$$\mathbb{E}[D^P_{n,m}D^P_{n,m'}] = \alpha^P_m\alpha^P_{m'} + \alpha^P_\ell.$$

Bounding the two occupancy indicators by this product gives

$$\operatorname{Var}(H^P_{n,y}) \le C\sum_{m,m'>y}\Lambda(m)\Lambda(m')\left(\frac{L^3}{mm'\ell} + \frac{L^2}{\ell^2}\right).$$

Apply Lemma 8.2. □

Lemma 14.3 (Transfer of the core variance)**.** *One has*

$$\operatorname{Var}(H^{\circ}_{n,y}) \le C_{\kappa,B_*} L(\log L)^{2B_*+1}.$$

Proof. The identification-graph proof of Lemma 9.5, with the present cutoff y, gives for every fixed k

$$\mathbb{E}(H^{\circ}_{n,y})^k + \mathbb{E}(H^{P}_{n,y})^k \le C_{\kappa,k,B_*} L^k (\log L)^{C_{k,B_*}}.$$

Indeed, a graph with $v \le 2k$ vertices contributes $L^v y^{-S}$ with $S \ge v/2$, and hence at most L^k times a fixed power of $\log L$.

Use Theorem 1.7 with an arbitrarily large exponent and truncate both corrections at L^2. Total variation compares their truncated first and second moments, while the fixed eighth moments make the tails $o(1)$. Thus the first two moments, and hence the variances, differ by $o(1)$. Apply Lemma 14.2. □

14.2 Adding the omitted submacroscopic cycles

Let

$$\mathcal{O}_n = \{1 \le r \le W : r \notin \mathcal{I}_n\}, \qquad W = n/8,$$

and

$$E_{n,m} = \sum_{\substack{r \in \mathcal{O}_n \\ m \mid r}} C_{n,r}.$$

Let $H^{\mathrm{sub}}_{n,y}$ be the large-prime correction formed from all cycles of length at most W.

Lemma 14.4 (Submacroscopic remainder variance)**.** *One has*

$$\operatorname{Var}(H^{\mathrm{sub}}_{n,y}) \le C_{\kappa,B_*} L(\log L)^{2B_*+1}.$$

Proof. For integers $a, b \ge 0$,

$$(a+b-1)_+ - (a-1)_+ \le ab + \frac{1}{2}(b)_2.$$

Hence

$$0 \le H^{\mathrm{sub}}_{n,y} - H^{\circ}_{n,y} \le J_1 + J_2,$$

where

$$J_1 = \sum_{m>y} \Lambda(m) D^{\circ}_{n,m} E_{n,m}, \qquad J_2 = \frac{1}{2} \sum_{m>y} \Lambda(m) (E_{n,m})_2.$$

Put

$$\alpha_m = \sum_{\substack{r \in \mathcal{I}_n \\ m \mid r}} \frac{1}{r} \le CL/m, \qquad \beta_m = \sum_{\substack{r \in \mathcal{O}_n \\ m \mid r}} \frac{1}{r} \le C \frac{\log L}{m}.$$

Every length is at most W, so the factorial-functional bound applies to all configurations of at most four selected cycles. Expanding J_1^2 by the equality pattern of its two core–omitted pairs gives, with $\ell = \operatorname{lcm}(m, m')$,

$$\mathbb{E} J_1^2 \le C_\kappa \sum_{m,m'>y} \Lambda(m)\Lambda(m') \big[\alpha_m \beta_m \alpha_{m'} \beta_{m'} + \beta_\ell \alpha_m \alpha_{m'} + \alpha_\ell \beta_m \beta_{m'} + \alpha_\ell \beta_\ell \big].$$

Similarly,

$$\mathbb{E}J_2^2 \le C_\kappa \sum_{m,m'>y} \Lambda(m)\Lambda(m') \left[\beta_m^2\beta_{m'}^2 + \beta_\ell\beta_m\beta_{m'} + \beta_\ell^2\right].$$

Using $\sum_{m>y}\Lambda(m)/m^2 = O(1/y)$ and Lemma 8.2, both expressions are bounded by a fixed power of $\log L$. This is absorbed by the right side of the asserted variance bound. Combine with Lemma 14.3. □

14.3 The macroscopic increment

Let $D_{n,m}^{\mathrm{mac}}$ count cycles of length greater than $W = n/8$, and define $H_{n,y}^{\mathrm{mac}}$ as the increase in $H_{n,y}$ when these cycles are added to the submacroscopic configuration.

Lemma 14.5 (Residual-gap harmonic sum)**.** *Let $T = b_n$, $W = n/8$, and $d \ge 1$. Uniformly in $N \le n$,*

$$\sum_{\substack{W<r\le N-T\\ d\mid r}} \frac{n}{r(N-r)} \le C\left(\frac{L}{d} + \frac{1}{T}\right).$$

Also

$$\sum_{\substack{r\le n\\ d\mid r}} \frac{1}{r} \le \frac{CL}{d}, \qquad \sum_{\substack{W<r\le n\\ d\mid r}} \frac{1}{r} \le \frac{C}{d}.$$

Proof. Since a contributing r is greater than $n/8$, one has $n/N \le 8$, and

$$\frac{n}{r(N-r)} \le C\left(\frac{1}{r} + \frac{1}{N-r}\right).$$

The first progression has bounded logarithmic range and mass $O(1/d)$. In the second put $t = N-r$. Then $t \ge T$ and lies in one residue class modulo d, so its mass is at most $1/T + (C/d)\log(n/T)$. The remaining assertions are ordinary harmonic summation. □

Lemma 14.6 (Macroscopic remainder variance)**.** *One has*

$$\mathrm{Var}(H_{n,y}^{\mathrm{mac}}) \le C_{\kappa,B_*} L(\log L)^{2B_*+1}.$$

Proof. There are at most seven cycles longer than W, and adding one cycle changes the correction by at most its logarithm. Hence

$$0 \le H_{n,y}^{\mathrm{mac}} \le 7L. \tag{80}$$

For

$$\omega_y(r,s) = \sum_{\substack{m>y\\ m\mid r,\ m\mid s}} \Lambda(m),$$

the elementary inequality used in the preceding proof gives $H_{n,y}^{\mathrm{mac}} \le J_{n,y}$, where, up to a fixed multiplicity,

$$J_{n,y} = \sum_{\substack{C\ne D\\ \max\{|C|,|D|\}>W}} \omega_y(|C|,|D|).$$

The square of $J_{n,y}$ contains at most four distinct cycle vertices.

Let $\mathcal{G}_n = \{K_n^\circ \geq 5\}$. Every monomial in $J_{n,y}^2$ selects at most four distinct cycles. On $\mathcal{G}_n$ at least one core cycle is therefore unselected; its length is at least $b_n = T$. Hence a selected length tuple with total R contributes on $\mathcal{G}_n$ only if $R \leq n - T$. More explicitly,

$$\mathbb{E}\left[\mathcal{C}_n(r_1, \ldots, r_q)\mathbf{1}_{\mathcal{G}_n}\right] \leq \mathbf{1}_{\{R \leq n-T\}} \frac{C_{\kappa,q} n}{r_1 \cdots r_q (n-R)}, \qquad q \leq 4. \tag{81}$$

Indeed, the left side vanishes when $R > n - T$; when $R \leq n - T$ we drop the indicator and apply the exact multi-clock identity and its first-moment bound.

Expand the two edge weights by prime powers $m, m' > y$ and put $\ell = \operatorname{lcm}(m, m')$. There are three identification graphs. We spell out the residual-gap summations, dropping only constraints whose removal increases the sums.

For the two-vertex graph the two edges are parallel, so both selected lengths are multiples of ℓ, and at least one is greater than W. By symmetry choose the first one to be macroscopic. Fixing the second length and applying Lemma 14.5 to the first gives

$$\mathcal{S}_2(m, m') \leq C \sum_{\substack{r_2 \leq n \\ \ell \mid r_2}} \frac{1}{r_2} \left(\frac{L}{\ell} + \frac{1}{T}\right) \leq C \left(\frac{L^2}{\ell^2} + \frac{L}{T\ell}\right).$$

For the three-vertex path, the vertex moduli are m, ℓ, m'. If the middle vertex is macroscopic, applying the residual-gap lemma there and ordinary harmonic bounds at the two outer vertices gives

$$\frac{L}{m}\frac{L}{m'}\left(\frac{L}{\ell} + \frac{1}{T}\right).$$

If the middle vertex is not macroscopic, both outer vertices must be macroscopic. Applying the residual-gap lemma to one outer vertex and the bounded-range macroscopic harmonic bound to the other gives the smaller quantity

$$C\frac{L}{\ell}\frac{1}{m'}\left(\frac{L}{m} + \frac{1}{T}\right)$$

(up to exchanging m, m'). Thus

$$\mathcal{S}_3(m, m') \leq C \left(\frac{L^3}{mm'\ell} + \frac{L^2}{Tmm'}\right).$$

For the four-vertex graph the two edges are disjoint and each has a macroscopic endpoint. Choose one macroscopic endpoint of each edge, apply the residual-gap lemma to the first, the bounded-range macroscopic harmonic bound to the second, and ordinary harmonic bounds to the two remaining vertices. Summing over the finitely many choices yields

$$\mathcal{S}_4(m, m') \leq C \left(\frac{L^3}{m^2 m'^2} + \frac{L^2}{T}\left(\frac{1}{mm'^2} + \frac{1}{m^2 m'}\right)\right).$$

Besides Lemma 8.2, use

$$\sum_{m > y} \frac{\Lambda(m)}{m^2} = O(1/y), \qquad \sum_{m \leq n} \frac{\Lambda(m)}{m} = O(L),$$

and

$$\sum_{m,m'\le n}\frac{\Lambda(m)\Lambda(m')}{\operatorname{lcm}(m,m')}=O(L^2).$$

For unequal prime bases the last sum factorizes into two copies of $\sum_{m\le n}\Lambda(m)/m$. For equal prime bases it is bounded by $C\sum_{p\le n}(\log p)^2/p=O(L^2)$, proving the last estimate. The three graph contributions are respectively bounded by

$$C\left(L^2\frac{\log y}{y}+\frac{L^3}{T}\right),$$
$$C\left(L^3\frac{\log y}{y^2}+\frac{L^4}{T}\right),$$
$$C\left(\frac{L^3}{y^2}+\frac{L^3}{Ty}\right).$$

Since $T=b_n\asymp L^3$ and $y=L/(\log L)^{B_*}$, their sum is at most $C_{\kappa,B_*}L(\log L)^{2B_*+1}$. Hence

$$\mathbb{E}[J_{n,y}^2\mathbf{1}_{\mathcal{G}_n}]\le C_{\kappa,B_*}L(\log L)^{2B_*+1}.$$

On $\mathcal{G}_n^c$, equations (80) and Corollary 11.2 give a contribution $O(L^{-1})$. This proves the result. □

Theorem 14.7 (Full centered arithmetic remainder)**.** *One has*

$$\operatorname{Var}(H_{n,y})\le C_{\kappa,B_*}L(\log L)^{2B_*+1}. \tag{82}$$

Proof. The full correction is the sum of the submacroscopic correction and its macroscopic increment. Apply $\operatorname{Var}(X+Y)\le 2\operatorname{Var}X+2\operatorname{Var}Y$ together with Lemmas 14.4 and 14.6. □

15 Final perturbation transfer

Proof of Theorem 1.9. By (77) and the definition of $D_{n,y}$,

$$\log O_n-m_n=(T_n^\sharp-\mathbb{E}T_n^\sharp)+\mathcal{R}_n,$$

where

$$\mathcal{R}_n=(D_{n,y}-\mathbb{E}D_{n,y})-(V_{n,y}-\mathbb{E}V_{n,y})-(H_{n,y}-\mathbb{E}H_{n,y}).$$

Put

$$u_n=L^{3/4}(\log L)^{B_*+2},\qquad \tau_n=Cy+2u_n.$$

Lemma 13.3 and Chebyshev give

$$\mathbb{P}\{|D_{n,y}^{\mathrm{sm}}-\mathbb{E}D_{n,y}^{\mathrm{sm}}|>u_n\}=o(L^{-1/2}).$$

Theorem 14.7 gives

$$\mathbb{P}\{|H_{n,y}-\mathbb{E}H_{n,y}|>u_n\}\le CL^{-1/2}(\log L)^{-3}=o(L^{-1/2}).$$

The bounded parts of Lemma 13.3 and (78) are absorbed by Cy. Therefore

$$\mathbb{P}\{|\mathcal{R}_n|>\tau_n\}=o(L^{-1/2}). \tag{83}$$

Moreover,

$$\delta_n := \tau_n/\sigma_{q,n} = o(L^{-1/2}),$$

because $y/\sigma_{q,n} = L^{-1/2}(\log L)^{-B_*}$ and $u_n/\sigma_{q,n} = L^{-3/4}(\log L)^{B_*+2}$.

Let F_n be the distribution function of $(T_n^\sharp - \mathbb{E}T_n^\sharp)/\sigma_{q,n}$. For every x,

$$F_n(x-\delta_n) - \mathbb{P}\{|\mathcal{R}_n| > \tau_n\} \le \mathbb{P}\left\{\frac{\log O_n - m_n}{\sigma_{q,n}} \le x\right\} \le F_n(x+\delta_n) + \mathbb{P}\{|\mathcal{R}_n| > \tau_n\}.$$

The Edgeworth approximants G_n have uniformly bounded derivatives. Proposition 13.2 and (83) therefore imply

$$\sup_x \left|\mathbb{P}\left\{\frac{\log O_n - m_n}{\sigma_{q,n}} \le x\right\} - G_n(x)\right| = o_\kappa(L^{-1/2}).$$

Use (70). Finally,

$$\frac{\sigma_{q,n}}{\sqrt{L^3/3}} = 1 + O\left(\frac{\log L}{L}\right) = 1 + o(L^{-1/2}),$$

and the normal and Edgeworth terms have bounded scale derivatives. This proves (1). □

Proof of Theorem 1.8. The Edgeworth correction is uniformly $O(L^{-1/2})$ and the remainder is $o(L^{-1/2})$. Apply Theorem 1.9. □

Proof of Corollary 1.10. Evaluate (1) at $x = 0$. Since $\varphi(0) = 1/\sqrt{2\pi}$, the absolute difference from $\Phi(0)$ is

$$\left(\frac{\sqrt{3}}{8\sqrt{2\pi}} + o_\kappa(1)\right) L^{-1/2}.$$

□

16 Scope, centering, and further questions

The endpoint and Edgeworth theorems use the exact mean m_n. The currently available estimate

$$|m_n - \tfrac{1}{2}L^2| = O_\kappa(L\log L)$$

is negligible for every $L^{-1/2+\varepsilon}$ rate but not at the sharp scale. An explicit deterministic center in Theorem 1.9 must approximate m_n to $o(L)$; its determination remains separate from normal approximation.

The fixed-moment hierarchy remains useful even after the endpoint theorem. It is derived from weaker square-root-depth input, provides independent checks at $k = 1, 2, 3$, and supplies the finite-order graph bounds used in the centered remainder proof. What fails at the endpoint is not fixed-order switching itself, but the generic L^k perturbation inequality applied before prime-power linearization.

Natural next questions include an explicit second-order center, a second Edgeworth term, a local limit theorem, and multivariate Edgeworth expansions for the cycle-count/order field.

Remark 16.1 (Status and dependency audit). The checkpoint hierarchy, polynomial passive-mark approximation, sharp Berry–Esseen theorem, and first Edgeworth expansion are proved from two source interfaces: the quantitative switching ODE used in the original proxy proof, and the free-depth finite-endpoint theorem of [7]. The latter is used only through Input Theorem 10.1; all Campbell–Stein, Fourier, linearization, graph, and perturbation arguments are contained in this supplement. No growing-order multi-clock estimate, residual-uniform factorial theorem, or macroscopic local law is assumed.

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